Enhancing decision reliability through a PSO-optimized hybrid multi-criteria decision-making framework

Shijie Sun [1], Xinyue Zhang [2,a], Jialin Li [3,a], Kejun Lu [4,a], Anna Qian [5,b], Zhanheng Sun [6,b], Mingren Zheng [7,b], Faan Chen [8,*]

[1] Business School, Monash University, Caulfield East VIC 3145, Australia

[2] Department of Industrial Engineering and Operations Research, Columbia University, New York, NY 10027, United States

[3] Harris School of Public Policy, University of Chicago, Chicago, IL 60637, United States

[4] Goizueta Business School, Emory University, Atlanta, GA 30322, United States

[5] School of Mathematical Sciences, University of Nottingham, University Park, Nottingham NG7 2RD, United Kingdom

[6] College of Letters & Science, University of Wisconsin–Madison, Madison, WI 53706, United States

[7] Tabor Academy, Marion, MA 02738, United States

[8] School of Engineering and Applied Sciences, Harvard University, Cambridge, MA 02138, United States

[a] Co-second authors of the article.

[b] Co-third authors of the article.

* Corresponding author.

E-mail address: faanchen@tongji.edu.cn (F. Chen).

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ABSTRACT

Effectively managing multi-criteria decision-making (MCDM) problems with reliability and robustness is crucial for any decision model, which is especially true for safety engineering. This study develops a hybrid MCDM model that integrates FANMA, measurement of alternatives and ranking according to compromise solution (MARCOS), and K-means clustering, termed FANMA–MARCOS–K-means, to create a robust and scalable decision support system. In particular, the model incorporates an artificial intelligence technique (i.e., particle swarm optimization) to improve the exploration of the search space during clustering, reduce the risk of the algorithm converging to local optima, and ensure global convergence. By integrating this approach, the model addresses a key limitation of the conventional K-means algorithm regarding the uncertainty in clustering outcomes owing to varying initial centroid selections. This significantly increases the stability and efficiency of decision-making, particularly in complex scenarios involving extensive datasets with high uncertainty. Extensive comparisons using a case study focused on transport safety engineering for the Organization of American States region demonstrated the adaptability, reliability, and robustness of the model, affirming its effectiveness in providing credible policy recommendations. Overall, this study offers decision-makers a reliable tool for navigating real-world MCDM activities, enabling the formulation of effective measures and action plans.

Keywords: Decision reliability; FANMA; MARCOS; K-means; Particle swarm optimization; Transport safety engineering

1 Introduction

Multi-criteria decision-making (MCDM) models are critical to solving real-world challenges, of which stability, reliability, and efficiency are essential for ensuring sound decision-making. This is particularly true for transport safety engineering. However, selecting the appropriate MCDM method, while considering the inherent sensitivity and uncertainty caused by input data variations, remains a challenge for decision-makers. Therefore, developing comprehensive MCDM models that can effectively mitigate these issues and maintain high levels of robustness, stability, and reliability is necessary.

In previous studies, extensive work has been conducted to refine the established MCDM methods that have demonstrated broad applicability in different domains and regions [1-7], Recent efforts have aimed to improve the adaptability and reliability of MCDM methods by integrating critical processes, such as aggregation and grouping. Such enhancements have contributed in increasing the robustness, decision accuracy, and effectiveness in complex decision-making contexts. Although these studies have led to the development

of more comprehensive MCDM systems, several research gaps remain, leading to five primary research motivations for this study: (1) The lack of universally applicable safety performance indicators (SPIs) hinders the generalizability of current models across different contexts. (2) Previous studies often focused on individual countries or smaller regions and neglected the broader inclusiveness and diversity required for cross-national applications, especially in regions with vast economic and geographical differences. This narrow focus resulted in methods that are difficult to apply beyond their original scope, potentially leading to incomplete decision-making outcomes. (3) Current methods often focus primarily on aggregating results and ranking alternatives, without considering the critical step of grouping similar objects for further development. (4) The effectiveness of MCDM models in complex scenarios is limited because important processes, such as aggregation and grouping, are not yet integrated. Existing methods also exhibit instability, especially when applied to small or medium-sized datasets, where uncertainty and decision outcomes are less predictable. This problem is exacerbated by the fact that these methods are sensitive to input variations, which can lead to inconsistent results. (5) Finally, the lack of studies on the deconstruction and decomposition of MCDM models makes it difficult to understand the underlying mechanisms and adaptability to different contexts. Therefore, an advanced MCDM system is required that integrates different national contexts, unifies weighting, aggregation, and grouping processes, and is sufficiently robust to deal with uncertainty and instability, especially in scenarios with limited data.

This study develops a hybrid decision model, FANMA–MARCOS–K-means with particle swarm optimization (PSO), to overcome the limitations of existing MCDM methods. This model integrates FANMA, measurement of alternatives and ranking according to compromise solution (MARCOS), and K-means clustering complemented by PSO. This model serves as a reliable tool for decision-making guidance and makes an important contribution to research and policymaking communities.

(1) First, a comprehensive set of SPIs was developed to execute transport safety decision-making within the member countries of the Organization of American States (OAS) and for potential applications in other regions.

(2) Second, the MCDM model performance was improved by integrating PSO with K-means clustering, which optimizes the initial centroid positions, reduces the risk of the algorithm converging to local optima, and ensures global convergence, thus overcoming the limitations of traditional clustering methods. In addition, the stability and accuracy of the clustering process was improved, especially in complex decision-making scenarios with high uncertainty. The integration of PSO and K-means reduces the dependency on the initial conditions, resulting in more consistent and reliable grouping outcomes. This robust and efficient model is suitable for large-scale datasets and diverse applications.

(3) Third, the proposed framework will enable policymakers in the OAS region to make informed decisions and implement strategic action plans to improve transport safety and achieve regional sustainable development goals.

2 Literature review

2.1 MCDM procedure

2.1.1 Uncertainty of the MCDM process

MCDM processes involve decision-making in complex scenarios with conflicting criteria and are subject to uncertainty due to the variability and imprecision of the input data. Researchers have investigated the application of uncertainty theories, such as those used in failure modes and effects analyses, to improve the robustness of decision-making models and account for the effects of input data variability on decision outcomes [8]. The importance of determining objective characteristics under uncertain conditions has been emphasized as crucial for improving MCDM performance [9].

Subjectivity in decision-making, influenced by individual preferences and biases, is another important source of inherent uncertainty [10], especially in complex and dynamic environments [11]. Recent studies have suggested data-driven approaches, such as distance learning, to reduce subjectivity by incorporating more objective data [10]. Determining the objective characteristics of MCDM methods is crucial for decision-making stability in dynamic decision environments with continuously evolving conditions [11].

Normalization techniques effectively deal with uncertainties in decision-making by standardizing diverse criteria and ensuring reliable comparisons between alternatives, despite the challenge of subjectivity, dynamic environments, and incomplete or unknown information, especially in decision criteria weighting [12].

Recent advances in MCDM methods have emphasized the importance of developing comprehensive evaluation systems and integrative approaches to enhance their applicability across various regions and sectors [13]. In particular, advanced techniques such as interval-valued Pythagorean fuzzy sets and gray stochastic methods improve the decision-making accuracy by incorporating fuzzy logic and regret theory [14, 15]. These methods have improved the reliability and robustness of MCDM models, especially in scenarios with a high degree of uncertainty.

In summary, dealing with uncertainty in MCDM processes requires a multi-faceted approach that includes sensitivity analysis, robustness testing, and advanced analytical techniques. These strategies can enhance the reliability and applicability of MCDM models in complex real-world scenarios.

2.1.2 Optimization of the MCDM procedure

Recent studies have provided insights into optimizing key stages of the MCDM procedure to improve the reliability and robustness of decision outcomes and thus the overall effectiveness of MCDM models [16, 17].

Data normalization is a crucial step in decision-making because it ensures consistent comparisons between criteria. It minimizes scale differences and improves accuracy and reliability [18]. This is particularly important when dealing with inconsistent or imprecise data inputs, including outliers [19]. Normalization provides a uniform basis for comparison

and ensures the integrity and reliability of MCDM outcomes in diverse decision-making scenarios.

Following normalization, assigning weights to the different criteria is crucial for decision making. The best–worst method provides a structured approach to derive optimal weighting by comparing the best and worst criteria [20]. Various weighting techniques can influence the prioritization and results. Accurate weighting ensures that the decision maker's preferences are correctly reflected, improving both the precision and robustness of the decision-making process [21].

Normalization and weighting of criteria are followed by ranking and aggregation of information for coherent decision-making. Parametric optimization techniques improve the consistency of the decision based on the chosen ranking model [22]. Different ranking models affect the robustness of MCDM outcomes [18] and ensure that decisions remain reliable and stable even with conflicting criteria.

The grouping of decision alternatives simplifies the decision-making process and improves the interpretability of results. In the final stage of the MCDM process alternatives are categorized based on shared performance characteristics, providing a clearer overview. This helps to identify trends and patterns, thereby contributing to more informed strategic decisions. Effective clustering ensures consistent evaluation of similar alternatives, increasing the overall robustness and reliability of the MCDM results.

In summary, optimizing the MCDM procedure involves a series of interconnected steps, including data normalization, weighting, ranking/aggregation, and grouping, which can improve the robustness, reliability, and applicability of MCDM models in complex decision-making contexts.

2.2 K-means applications and limitations

K-means clustering is an unsupervised machine learning algorithm that partitions datasets into k clusters and represents each cluster by the centroid of each cluster using its data points. It aims to minimize variance by ensuring that data points are close to their centroids.

K-means is a versatile algorithm used in various fields such as Earth sciences, mineral prospectivity mapping, structural damage identification [23], and ambient intelligence. It has been integrated into hybrid algorithms for data-driven modeling, and used for processed large-scale environmental datasets [24, 25]. In addition, K-means has been used for nonparametric color image segmentation in image processing tasks and has demonstrated its versatility in prioritizing health, safety, and environmental risks [26, 27].

Despite its efficiency, K-means has several limitations. These include sensitivity to initial centroid placement [28], which can lead to inconsistent clustering results [29], and scalability and computational complexity [30] issues for large datasets as the computational complexity increases with dataset size [31]. Moreover, K-means assumes spherical, equally sized clusters, which can hinder performance on real-world data with irregular shapes [32]. Finally, the algorithm requires advanced cluster number

specification, which is challenging without prior knowledge [27], and is sensitive to noise and outliers, which can distort the clustering results [33].

Modifications to address these limitations include combining K-means with harmony search and artificial bee colony algorithms to improve stability and convergence in large-scale data clustering [24], and utilizing adaptive unsupervised learning techniques to improve performance in nonparametric settings [26]. Moreover, integrating principal component analysis (PCA) with K-means reduces instability and improves clustering accuracy through dimensionality reduction [32]. Hybrid models that combine K-means with evolutionary algorithms [29] and an improved K-means algorithm focusing on enhanced initialization and iterative refinement processes [33] have been developed to improve algorithm efficiency and clustering consistency.

In summary, although K-means has limitations, advances such as improved initialization methods, adaptive learning, dimensionality reduction, and hybrid models have improved its robustness and applicability and ensured its relevance in various domains and applications.

2.3 Institutional compacity of the OAS in transport safety management

The OAS has adopted a comprehensive strategy to improve transport safety by implementing tailored frameworks and policies at various levels, including broad area, subregional, and national levels. This strategy accounts for the different contexts and needs of its member states, and it highlights the efforts and insights of various studies.

The OAS supports initiatives aimed at saving lives through improved transport safety measures. Hughes [34] developed a comprehensive framework for transport safety strategies, including policy implementation, infrastructure development, and behavioral interventions to reduce traffic-related fatalities and injuries. This underscores the need for a coordinated approach to transport safety management.

At the sub-regional level, Martinez's [35] study on transport safety issues in Latin America and the Caribbean emphasizes the need for tailored strategies that take into account the socio-economic and infrastructural diversity of these subregions. This research underscores the significance of customized approaches in addressing unique challenges and opportunities within the OAS.

At the national level, a study by Egilmez and McAvoy [36] analyzed the transport safety performance across US states using a DEA-based Malmquist productivity index. The research offers insights into the effectiveness of safety measures and policies at the state level and provides valuable lessons for other OAS countries. The findings can help identify best practices and areas for improvement in national transport safety management. Various countries within the OAS have been benchmarked for their transport safety performance. In addition, a study by Brubacher [37] on the effects of increased speed limits on rural highways in British Columbia, Canada, underscores the importance of evidence-based policy making in improving national transport safety strategies and emphasizes the influence of legislative changes. Sagaris' study [38] on citizen participation in Santiago, Chile, emphasizes the importance of local engagement

and community involvement in improving transport safety and promoting sustainable urban transportation solutions at the national level.

In summary, the multilayered approach to transport safety management developed by the OAS aims to create a comprehensive transport safety framework that can significantly reduce traffic-related fatalities and injuries throughout the Americas.

2.4 Research gaps

Existing studies on MCDM and transport safety management in the OAS region reveal significant research gaps in methodological and practical aspects.

2.4.1 Methodological

Aggregation versus grouping: Existing MCDM models focus primarily on aggregating results to rank alternatives. However, there is a lack of comprehensive approaches that integrate aggregation and grouping as a unit. This gap highlights the need for models that rank alternatives and group them for targeted strategy formulation.

Instability in small to medium sample sizes: The instability of MCDM methods is evident in scenarios with small to medium sample sizes, leading to significant variations in decision outcomes and undermining the robustness of the decision-making process. Therefore, MCDM methods that remain stable and reliable even with limited data are needed.

2.4.2 Practical

Lack of universally applicable evaluation metrics systems: The current literature lacks universally applicable evaluation metrics systems for transport safety development. This is because most studies focus on specific contexts or regions, which limits their applicability across different settings and poses a challenge for cross-national comparisons and in formulating comprehensive transport safety strategies.

Focus on single countries or small regions: Previous transport safety studies focused primarily on individual countries or regions, lacking the inclusivity and diversity required for cross-national applicability. This narrow focus restricts the ability to draw comprehensive conclusions for larger, diverse regions, such as the OAS. Broader studies encompassing multiple countries and regions are needed for a more comprehensive transport safety policy.

Scarcity of studies on deconstruction and decomposition: Few studies investigated the deconstruction and decomposition of MCDM models, which can provide deeper insights into decision-making processes and increase transparency. This suggests that more research is needed on decomposing MCDM processes into their constituent parts to improve clarity and effectiveness.

Addressing these research gaps is critical for improving transport safety strategies in the OAS region. Future research should focus on developing robust, integrative models to help decision-makers in developing sustainable transport safety policies.

3 Methodology

3.1 FANMA

The FANMA method is a robust MCDM approach that uses the distance to the ideal point and early weight normalization [39] to objectively determine the criteria weights and minimize the influence of decision maker subjectivity. First, the decision matrix is normalized and the distance between each criterion and the ideal point is calculated, which represents the best possible values for each criterion.

Next, the decision matrix is normalized, and the sum of squares for each criterion is calculated. This gives the distance to the ideal point, and weights are calculated based on the inverse of these sums. Criteria that are closer to the ideal point are given more weight. The weights are then normalized to one to ensure a balanced distribution. FANMA offers several advantages over conventional weighting methods, such as criteria importance through inter-criteria correlation (CRITIC) and Entropy. It relies on objective measures such as contrast intensity and standard deviation, thus avoiding subjective biases. CRITIC and Entropy aim to determine objective weightings but differ in their approach. CRITIC accounts for the correlations between criteria [40] and gives a lower weight to highly correlated criteria, whereas Entropy measures the information provided by each criterion [41]; however, it can sometimes overemphasize a less important criterion with high variability.

FANMA uses an additive normalization method to ensure the consistency of the scalarization process. It considers the type of each criterion (maximization or minimization) and standardizes the maximum value of the criterion across all criteria, providing a straightforward and consistent approach to normalization.

FANMA balances the weight distribution by considering inherent information and conflicts between the criteria and emphasizes their importance. In contrast, CRITIC may not fully account for internal information dynamics as it focuses on the relationships between criteria. Entropy, while useful, may not always accurately reflect the decision-making context as it focuses on variability.

In summary, FANMA's structured weight determination method and ability to mitigate subjective biases make it a powerful tool for multi-criteria decision-making. It ensures a balanced representation of all criteria and leads to more reliable and objective results and is therefore crucial for the development of robust and effective MCDM methods.

3.2 MARCOS

The MARCOS method is an MCDM technique that ranks and aggregates alternatives using both ideal and anti-ideal solutions. It was developed in [42] and involves creating an initial decision matrix with decision values for each alternative based on various criteria. The ideal solution includes the maximum and minimum values for the benefit and cost criteria, whereas the anti-ideal solution includes the minimum and maximum values for both.

7

In the next step, the extended decision matrix is normalized by dividing each decision value by the ideal and anti-ideal solution for the benefit and cost criteria so that all criteria are comparable. The decision values are then multiplied by their respective criteria weights to form a weighted matrix that reflects the relative importance of each criterion.

The MARCOS method calculates the degrees of utility for each alternative by comparing their normalized and weighted values with the ideal and anti-ideal solutions. The utility function, derived from these degrees, measures the overall performance of each alternative, taking into account its strengths and weaknesses. Finally, the alternatives are ranked based on their utility function values, providing an objective basis for decision-making. The comprehensive evaluation, objective aggregation, and robust ranking of this method make it a valuable tool for complex decision-making scenarios.

When compared with other aggregation methods, such as the technique of ordering preference by similarity to ideal solution (TOPSIS) and VlseKriterijumska optimizacija I Kompromisno Resenje (VIKOR), several distinctions become apparent. TOPSIS ranks alternatives based on their geometric distance to ideal and anti-ideal solutions [43] and aims to find the shortest and most distant alternatives. However, it can be sensitive to the criteria scales and cannot always deal with conflicting criteria. VIKOR ranks alternatives by identifying the one that is closest to the ideal solution [44] and considers a compromise solution to minimize the decision maker's regret. VIKOR aims to find a solution that is acceptable to the decision-maker and stakeholders, emphasizing compromise solutions and pre-specified criteria weights that can lead to subjective biases.

The MARCOS method provides a nuanced, balanced evaluation of alternatives by incorporating both ideal and anti-ideal solutions. The normalization process ensures comparable criteria and thus reduces potential biases. In addition, MARCOS's utility function offers a comprehensive assessment of alternatives, accounting for strengths and weaknesses, making it a valuable alternative to TOPSIS and VIKOR. This method is particularly useful in complex decision-making scenarios and provides a balanced, objective, and thorough methodology for MCDM.

3.3 K-means enhancement with PSO

The K-means algorithm is an unsupervised machine learning algorithm used to partition a dataset into K clusters [33], where each data point belongs to the cluster with the nearest mean. It involves randomly initializing the k centroids, assigning each data point to the nearest centroid, and recalculating the centroids as the mean of all the points assigned to them. This process is repeated iteratively until the centroids no longer change significantly or a predetermined number of iterations is reached. Despite its simplicity and efficiency, K-means has notable limitations, such as sensitivity to the initial placement of centroids, difficulty in determining the optimal number of clusters, and a tendency to converge to the local minima. An improved version of K-means integrated with PSO has been developed to address these limitations. PSO is an optimization technique that iteratively improves the candidate solutions based on a fitness function [45], inspired by

bird flocking and fish schooling. It helps in determining better initial centroids in K-means clustering and improves convergence properties and algorithm performance.

By integrating PSO with K-means, the positions of the centroids are optimized, which enables more effective exploration of the search space and avoids local minima. This hybrid approach uses a swarm of particles that represents potential solutions, which are adjusted based on the fitness function. This improves the accuracy of the K-means algorithm and presents a more global optimal solution than that achieved using conventional methods. This integration leads to better stability, efficiency, and effectiveness in clustering and is therefore well-suited for complex and high-dimensional datasets.

Hierarchical cluster analysis (HCA) is a clustering method that builds a hierarchy of clusters by merging smaller clusters into larger clusters (agglomerative) or dividing larger clusters into smaller clusters (divisive) [45]. In contrast to K-means, HCA does not require pre-specified cluster numbers. However, HCA can be computationally intensive and has scalability problems.

The K-means with PSO method is a more efficient and scalable alternative to HCA, which may not work well with large or high-dimensional datasets. The hybrid PSO-K-means method leverages the PSO optimization capabilities to improve the clustering process. It provides more robust and reliable results, making it ideal for complex datasets where conventional methods may not suffice.

3.4 Innovative FANMA–MARCOS–K-means model with PSO

This study proposes a novel model that integrates weighting, ranking, and grouping into a unified framework, termed the FANMA–MARCOS–K-means model with PSO. FANMA was used for indicator weighting, MARCOS was adopted for ranking, and K-means clustering was enhanced with PSO to optimize the grouping process.

Step 1. Decision matrix construction

The first step is to construct a decision matrix A, in which each element A_{ij} represents the performance value of the ith alternative evaluated with respect to the jth criterion. M represents the number of alternatives, whereas n represents the number of criteria. The decision matrix A is represented as follows:

$$A = \left[a_{ij} \right]_{m \times n} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \quad (1)$$

Step 2. Decision matrix normalization using the min-max method

The decision matrix A was normalized using the min-max normalization method to ensure the comparability of different criteria. This process transforms the original data a_{ij}

9

to bring all criteria to a common scale. The normalization differed for the cost and benefit criteria as follows:

Cost:

$$r_{ij} = \frac{a_{j\max} - a_{ij}}{a_{j\max} - a_{j\min}} \tag{2}$$

Benefit:

$$r_{ij} = \frac{a_{ij} - a_{j\max}}{a_{j\max} - a_{j\min}} \tag{3}$$

where a_{ij} represents the original value of the ith alternative for the jth criterion, a_{jmax} denotes the maximum value for criterion j, and a_{jmin} represents the minimum value for criterion j.

After normalization, the decision matrix transforms into the normalized matrix R, as shown in Eq. (4):

$$R = \left[r_{ij} \right]_{m \times n} = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ r_{21} & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ r_{m1} & r_{m2} & \cdots & r_{mn} \end{bmatrix} \tag{4}$$

Step 3. Calculating the weights for each criterion using FANMA.

The weight w_j for each criterion is determined by the following optimization model after normalization. The assignment of weights aims to minimize the distance between each alternative and the "ideal" alternative, where w_j denotes the weight for the j-th criterion.

(1) Ideal alternative

For each criterion j, the ideal value y_j^* is defined as the maximum of the normalized value r_{ij} from Eq. (4) for criterion j among all m alternatives in Eq. (5):

$$y_j^* = \max(r_{1j}, r_{2j}, ..., r_{mj}) \tag{5}$$

(2) Distance from the ideal alternative

The quadratic distance g_i between the performance of the ith alternative and the ideal alternative is expressed by Eq. (6), where w_j denotes the weighting for the jth criterion and r_{ij} represents the normalized performance value for the ith alternative and jth criterion:

$$g_i = \sum_{j=1}^{n} w_j \times (y_j^* - r_{ij})^2 \tag{6}$$

(3) Optimization model and output

The optimization model consists of an objective function and constraints defined in Eq. (7) and Eq. (8):

Objective Function: The total distance G was minimized to determine the optimal weight, which is the sum of the distances g_i for all alternatives.

$$\min G = \sum_{i=1}^{m} g_i = \sum_{i=1}^{m} \sum_{j=1}^{n} w_j \times (y_j^* - r_{ij})^2 \tag{7}$$

Constraints: The weights w_j were subject to the following constraints: the sum of the weights of each criterion must be 1, and each weight must be non-negative.

$$\sum_{j=1}^{n} w_j = 1$$
$$w_j \geq 0, \forall j \tag{8}$$

Solution for the Weights: Using an optimization method (e.g., Lagrange multipliers), the weights w_j for each criterion were calculated using Eq. (9):

$$w_j^* = \frac{\sum_{i=1}^{m} (y_j^* - r_{ij})^2}{\sum_{j=1}^{n} \sum_{i=1}^{m} (y_j^* - r_{ij})^2} \tag{9}$$

Step 4. Calculating the aggregated scores for the alternatives and creating the rankings using MARCOS.

After calculating the weight w_j for each criterion in Eq. (9), the alternatives were ranked using the MARCOS method, wherein utility degrees and ranking scores were calculated by comparing the alternatives with the ideal and anti-ideal solutions.

(1) Definition of ideal and anti-ideal solutions

Ideal solution (AI): For each criterion j, the ideal value v_j^{AI} is defined as the best (maximum) performance among all alternatives for the beneficial criteria and worst (minimum) for the cost criteria. The ideal value is defined using Eq. (10).

$$v_j^{AI} = \begin{cases} \max(r_{1j}, r_{2j}, ..., r_{mj}) \text{ for benefit criteria} \\ \min(r_{1j}, r_{2j}, ..., r_{mj}) \text{ for cost criteria} \end{cases} \tag{10}$$

Anti-ideal solution (AAI): Conversely, the anti-ideal value v_j^{AAI} for each criterion j has the worst (minimum) performance among all alternatives for the beneficial criteria and the best (maximum) performance for the cost criteria. The anti-ideal value is defined using Eq. (11).

$$v_j^{AAI} = \begin{cases} \min(r_{1j}, r_{2j}, ..., r_{mj}) \text{ for benefit criteria} \\ \max(r_{1j}, r_{2j}, ..., r_{mj}) \text{ for cost criteria} \end{cases} \tag{11}$$

(2) Computation of aggregated scores for each alternative

The aggregated score for alternative i (S_i) represents the weighted sum of the normalized values for alternative i in Eq. (12), where w_j denotes the weight of criterion j

(from Eq. (9)), and r_{ij} represents the normalized value for alternative i with respect to criterion j (from Eq. (4)):

$$S_i = \sum_{j=1}^{n} w_j \times r_{ij} \tag{12}$$

The aggregated score for the ideal solution (S_{AI}) denotes the weighted sum of the values for the ideal solution in Eq. (13), where v_j^{AI} denotes the ideal solution for criterion j.

$$S_{AI} = \sum_{j=1}^{n} w_j \times v_j^{AI} \tag{13}$$

The aggregated score for the anti-ideal solution (S_{AAI}) denotes the weighted sum of the anti-ideal solution values in Eq. (14), where v_j^{AAI} represents the anti-ideal solution value for criterion j.

$$S_{AAI} = \sum_{j=1}^{n} w_j \times v_j^{AAI} \tag{14}$$

(3) Utility degree calculation

Upon obtaining the aggregated scores, the utility degree of each alternative i can be calculated with respect to both the ideal and the anti-ideal solution.

Utility degree with respect to the ideal solution: K_i^+ denotes the ratio of the aggregated score for alternative i to that for the ideal solution in Eq. (15), where S_i represents the aggregated score for alternative i, and S_{AI} denotes the aggregated score for the ideal solution:

$$K_i^+ = \frac{S_i}{S_{AI}} \tag{15}$$

Utility degree with respect to the anti-ideal solution: K_i^- denotes the ratio of the aggregated score for alternative i to that for the anti-ideal solution in Eq. (16), where S_i represents the aggregated score for alternative i, and S_{AAI} denotes the aggregated score for the anti-ideal solution.

$$K_i^- = \frac{S_i}{S_{AAI}} \tag{16}$$

(4) Utility function and ranking

The utility functions for positive $f(K_i^+)$ and negative $f(K_i^-)$ are denoted by the corresponding utility degrees from Eq. (15) and Eq. (16) into the sum of the total degrees of utility in Eq. (17).

$$f(K_i^+) = \frac{K_i^+}{K_i^+ + K_i^-}$$

$$f(K_i^-) = \frac{K_i^-}{K_i^+ + K_i^-} \tag{17}$$

12

The utility function $f(K_i)$ for each alternative i is a compromise of the observed alternative in relation to the ideal and anti-ideal solutions. The utility function of the alternatives is defined using Eq. (18):

$$f(K_i) = \frac{K_i^+ + K_i^-}{1 + \frac{1 - f(K_i^+)}{f(K_i^+)} + \frac{1 - f(K_i^-)}{f(K_i^-)}} \tag{18}$$

Finally, the alternatives are ranked based on the utility function values $f(K_i)$, with higher values representing better alternatives.

Step 5. Grouping the alternatives with the PSO-enhanced K-means model

The alternatives are then grouped into clusters using the PSO-enhanced K-means model. PSO optimizes the initial positions of the centroids in K-means clustering, improving the convergence properties and overall performance of the grouping process.

(1) Optimization of the PSO algorithm

First, a swarm of particles is initialized. Each particle represents a potential solution to the clustering problem, which corresponds to a set of centroids for k clusters.

The initial positions of the particles are randomly selected centroids within the bounds of data matrix R, which represents the normalized decision matrix. The position of each particle P_i is defined by a set of k centroids, as shown in Eq. (19), where C_i denotes the centroid of cluster i and k represents the number of clusters.

$$C = \{C_1, C_2, ..., C_k\} \tag{19}$$

Each centroid C_i has a dimensionality equal to the number of criteria used in the decision matrix.

(2) PSO iterative process

PSO optimizes the centroids by iteratively improving the particle positions based on a fitness function, known as the within-cluster sum of squares (WCSS) of K-means clustering.

Fitness function for PSO

The fitness function for PSO is designed to evaluate the quality of the current centroids by calculating the compactness of the clusters. This is usually done by minimizing the WCSS, as defined in Eq. (20).

$$g(C) = \sum_{i=1}^{k} \sum_{r_{ij} \in Q_k} \left\| r_{ij} - C_k \right\|^2 \tag{20}$$

where C_k represents the centroid of cluster k in Eq. (19), Q_k represents the set of data points assigned to cluster k, r_{ij} represents the normalized value of the ith alternative with respect to the jth criterion in the normalized decision matrix R from Eq. (4), and k denotes the number of clusters:

Velocity update:

The velocity of each particle is updated as follows:

$$v_p^{(t+1)} = w v_p^{(t)} + c_1 r_1 \times (P_p^* - x_p^{(t)}) + c_2 r_2 \times (g^* - x_p^{(t)}) \quad (21)$$

where

- $x_p^{(t)}$ represents the current position (centroid configuration) for particle p.
- $v_p^{(t)}$ represents the velocity of particle p at iteration t.
- P_p^* represents the best previous position (centroid configuration) of particle p.
- g^* represents the global best position for all particles.
- w represents the inertial weighting that controls the influence of the previous velocity.
- c_1 and c_2 represent the cognitive and social coefficients that influence the attraction towards the best-known position of the particle and the global best position, respectively.
- r_1 and r_2 represent random values between 0 and 1 that introduce stochastic behavior into the velocity update.

Position update:

The position of each particle (centroid) is updated as follows:

$$v_p^{(t+1)} = x_p^{(t)} + x_p^{(t+1)} \quad (22)$$

where

- $x_p^{(t)}$ represents the position (centroid configuration) of particle p in iteration t.
- $x_p^{(t+1)}$ represents the position (centroid configuration) of particle p in iteration t+1.
- $v_p^{(t+1)}$ represents the velocity of particle p in iteration t+1.

Fitness evaluation:

After each update, the PSO algorithm assesses the new centroid positions using the fitness function $g(C)$. If the fitness of a particle improves, its personal best position, P_p^* is updated. Similarly, if the particle has the best fitness in the entire swarm, the global best position $g*$ is updated.

(3) K-means clustering with optimized centroids

The k-means algorithm was applied to the data points using the optimized cluster centers obtained from the PSO process, where each data point was assigned to the nearest cluster center and the cluster centers were iteratively updated until convergence.

The optimized centroids $C_1, C_2, \ldots, C_k$ were used as starting centroids for K-means clustering. K-means assigned each data point to the nearest centroid C_k in Eq. (23):

$$Cluster(i) = \arg\min_k \left\| r_{ij} - C_k \right\| \quad (23)$$

where r_{ij} represents the normalized value of the ith alternative with respect to the jth criterion, C_k denotes the centroid of cluster k, and $Cluster(i)$ indicates that the ith alternative is assigned to cluster k with the nearest centroid.

(4) Stopping criteria for K-means

The K-means algorithm terminates the iterations if a convergence criterion is satisfied:

14

Convergence by inertia: The algorithm stops when the change in the WCSS between successive iterations is less than the predefined convergence threshold ε. Typically, K-means converges when

$$\left|W^{(t+1)} - W^{(t)}\right| \leq \varepsilon \tag{24}$$

where $W^{(t)}$ denotes WCSS (sum of squared distances between the data points and their assigned centroids) in iteration t, ε represents the convergence tolerance, and t denotes the current iteration index.

This ensures that the K-means algorithm stops when the change in the WCSS is minimal. This means that the centroids have stabilized, and further iterations will not bring any significant improvements.

Convergence through maximum iterations: If the algorithm fails to converge by inertia within the set maximum number of iterations, it generally stops after reaching the limit T_{max}:

$$t \geq T_{\max} \tag{25}$$

where T_{max} represents the maximum number of iterations predefined for the model, and t represents the current iteration index.

This serves as a safeguard to prevent the algorithm from running indefinitely when convergence is slow.

4 Results and discussion

4.1 Case study

4.1.1 Index construction

SPIs are essential for measuring transport safety and guiding strategic policies within the OAS region. They provide a comprehensive framework for assessing transport safety performance and facilitate benchmarking between countries. This enables policymakers to identify strengths and weaknesses, prioritize improvements, and take targeted actions, promoting a collaborative approach to improving transport safety.

The selection of SPIs for transport safety is based on a comprehensive literature review and a systematic approach based on the "Human-Vehicle-Road-Environment-Management" system and the "Pre-event, Event, Post-event" framework. This process ensures that the indicators cover all critical aspects of transport safety and are robust and meaningful for the assessment of transport safety performance. The chosen SPIs serve as vital tools for policymakers and help to develop effective strategies to improve transport safety in the OAS region, as shown in Fig. 1.

15

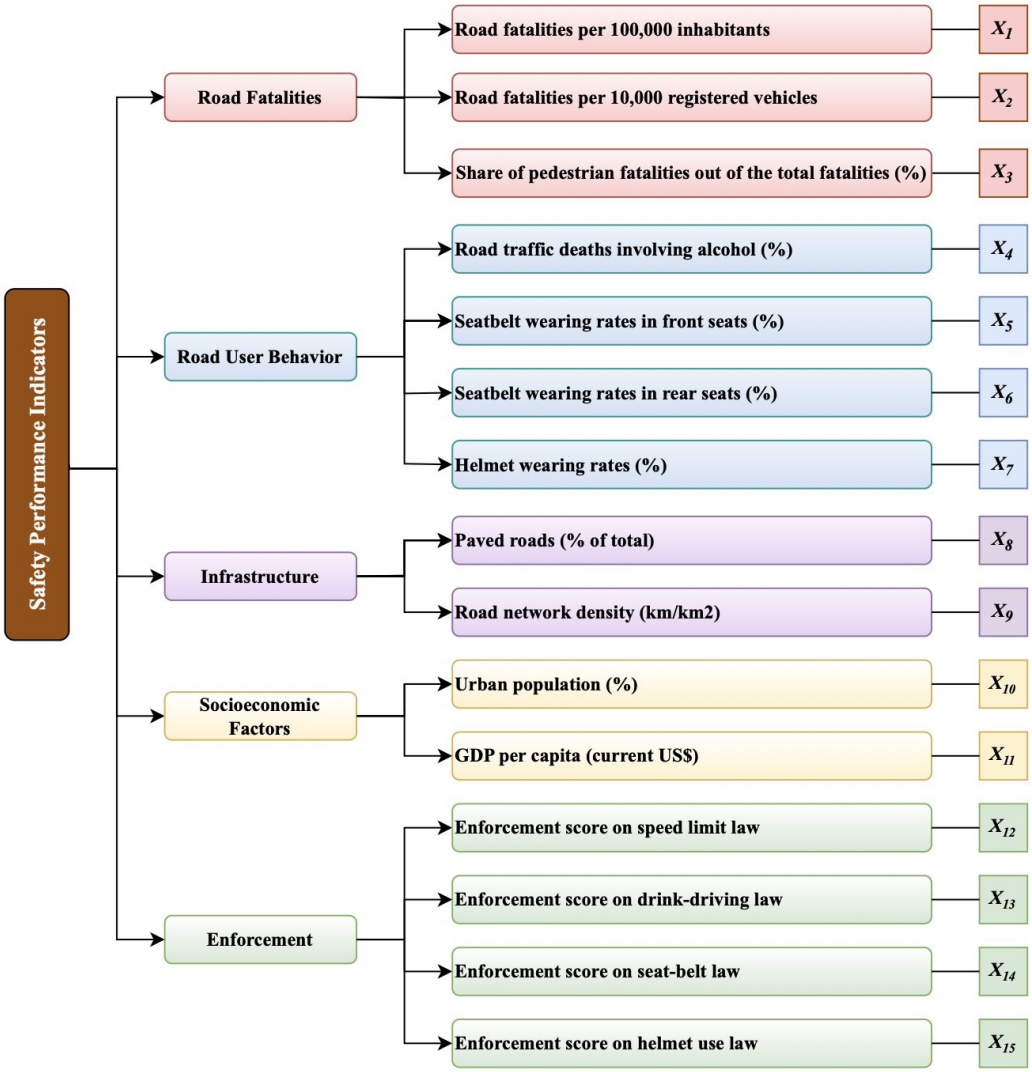


Fig. 1 SPIs used in this study.

4.1.2 Data sources

Data on the set of SPIs (years 2010 and 2020) were collected from 35 OAS countries. Indicators A1, B1, and B2 are from the World Health Organization (WHO) Global Health Observatory [46]. Indicator A2 was calculated using data provided by the WHO on the number of road fatalities and registered motor vehicles [46]. Indicator C1 was taken from Wikipedia [47] and ChartsBin [48]. C2 was calculated based on road network length and land area data reported by Statista [49], ChartsBin [48], and the World Bank [50, 51]. Indicators D1 and D2 were extracted from the World Bank database [50, 51]. Indicators B3, B4, E1, E2, E3, and E4 were derived from the global status report on road safety [52-55]. The human development index (HDI) was taken from the annual human development report published by the United Nations Development Programme [56, 57].

16

4.1.3 Data processing

The data processing phase involved careful and methodical handling of outliers and missing values. Statistical methods were used to detect and eliminate outliers to ensure the integrity and accuracy of the data analysis. Outliers were carefully examined and excluded if deemed erroneous or unrepresentative.

The study used a systematic approach to fill gaps in the dataset by replacing missing values with data from previous subsequent years. This imputation method ensures that missing values are replaced with reasonable and contextually relevant estimates so that the continuity and consistency of the dataset is maintained. When data from adjacent years were unavailable, interpolation techniques were used to estimate missing values while preserving the overall structure of the data and avoiding significant bias.

4.2 Computational results

4.2.1 Ranking

A comparative analysis of the scores and rankings of OAS countries for 2010 and 2020 is presented in Table 1, highlighting key trends and characteristics.

Table 1 Safety scores and rankings of OAS countries.

Country	ISO	2010		2020	
		Score	Rank	Score	Rank
Antigua and Barbuda	AG	0.7810	10	0.7824	14
Argentina	AR	0.6098	21	0.7642	16
Barbados	BB	0.8485	4	0.8558	9
Belize	BZ	0.5246	31	0.6891	23
Bolivia	BO	0.4753	32	0.5255	33
Brazil	BR	0.7616	11	0.9215	2
Canada	CA	1.0000	1	1.0000	1
Chile	CL	0.6635	17	0.8840	6
Colombia	CO	0.7415	12	0.7200	22
Costa Rica	CR	0.8178	7	0.7401	19
Cuba	CU	0.7817	9	0.7642	15
Dominica	DM	0.7030	14	0.5474	30
Dominican Republic	DO	0.6022	24	0.4659	34
Ecuador	EC	0.4180	33	0.7594	17
El Salvador	SV	0.5532	27	0.6477	26
Grenada	GD	0.7315	13	0.6850	24
Guatemala	GT	0.5775	26	0.6014	27
Guyana	GY	0.0000	35	0.7365	20
Haiti	HT	0.4069	34	0.0000	35
Honduras	HN	0.6062	22	0.7566	18
Jamaica	JM	0.6037	23	0.5463	31
Mexico	MX	0.5526	28	0.7949	12
Nicaragua	NI	0.5292	30	0.8480	10
Panama	PA	0.8440	5	0.7347	21
Paraguay	PY	0.5886	25	0.6592	25
Peru	PE	0.6111	20	0.5980	28
Saint Lucia	LC	0.6223	19	0.5319	32
Saint Vincent and the Grenadines	VC	0.9656	2	0.9035	4

17

St. Kitts and Nevis	KN	0.8154	8	0.7926	13
Suriname	SR	0.8363	6	0.8261	11
The Bahamas	BS	0.6720	16	0.8813	7
Trinidad and Tobago	TT	0.6788	15	0.8687	8
United States	US	0.8907	3	0.9138	3
Uruguay	UY	0.5522	29	0.8961	5
Venezuela	VE	0.6379	18	0.5699	29

As shown in Table 1, Canada consistently ranks first in the OAS, with a perfect score of 1.0000. The United States ranked third in 2010 and again in 2020, with scores of 0.8907 and 0.9138, respectively. Saint Vincent and the Grenadines moved up from second to fourth place, with scores of 0.9656 and 0.9035, respectively.

Over the past decade, several countries have made significant improvements in their transport safety performance. Brazil rose from 11th place in 2010 to second place in 2020 with scores of 0.7616 and 0.9215, respectively. Chile has improved significantly, moving up from 17th place in 2010 (0.6635) to sixth place in 2020 (0.8840). Ecuador has made considerable progress, moving up from 33rd place in 2010 (0.4180) to 17th place in 2020 (0.7594).

Conversely, Guyana and Haiti have consistently ranked lower, with Guyana in last place in 2010 and Haiti in 35th place in 2020. Guyana's score of 0.0000 remained at the bottom, whereas Haiti's score of 0.4069 ranked 34th. Both countries have made minimal progress.

4.2.2 Grouping

A comparative analysis of the grouping of OAS countries' transport safety performance between 2010 and 2020, showing significant trends and shifts in country groups, is listed in Table 2.

Table 2 Safety scores and groups of OAS countries.

Country	ISO	2010		2020	
		Score	Group	Score	Group
Antigua and Barbuda	AG	0.7032	2	0.6592	3
Argentina	AR	0.3544	2	0.5561	4
Barbados	BB	0.6515	2	0.6535	3
Belize	BZ	0.7418	4	0.8269	4
Bolivia	BO	0.6925	4	0.7549	4
Brazil	BR	0.5151	2	0.7631	4
Canada	CA	0.9469	1	0.0000	1
Chile	CL	0.6704	2	0.5843	3
Colombia	CO	0.6685	4	0.8272	4
Costa Rica	CR	0.3481	4	0.4505	3
Cuba	CU	0.0833	2	0.8274	4
Dominica	DM	0.5616	4	0.7471	4
Dominican Republic	DO	0.7358	4	0.7313	4
Ecuador	EC	0.7609	4	0.8191	4
El Salvador	SV	0.7279	4	0.7879	4
Grenada	GD	0.6267	4	0.5667	4
Guatemala	GT	0.7234	4	0.8139	4
Guyana	GY	0.7597	4	0.7583	4

18

Haiti	HT	0.6580	4	0.6759	4
Honduras	HN	0.6906	4	0.7252	4
Jamaica	JM	0.7599	4	0.8222	4
Mexico	MX	0.0414	2	0.5231	4
Nicaragua	NI	0.6731	4	0.7034	4
Panama	PA	0.3549	4	0.5244	3
Paraguay	PY	0.7610	4	0.8267	4
Peru	PE	0.7551	4	0.8033	4
Saint Lucia	LC	0.1505	4	0.5630	4
Saint Vincent and the Grenadines	VC	0.6478	4	0.5841	4
St. Kitts and Nevis	KN	0.6589	2	0.6121	3
Suriname	SR	0.3889	4	0.8198	4
The Bahamas	BS	0.0000	3	0.4707	3
Trinidad and Tobago	TT	0.7062	2	0.6353	3
United States	US	0.9495	1	0.0000	2
Uruguay	UY	0.7080	2	0.4245	3
Venezuela	VE	0.7099	2	0.8270	4

As shown in Table 2, Canada consistently ranks high in transport safety, remaining in Group 1 for both 2010 and 2020, whereas the United States moves up from Group 1 to Group 2 in 2020, indicating a relative decline in performance compared to the other top-performing countries over the past decade.

Conversely, Haiti and Guyana have consistently performed poorly in improving transport safety, ranking in Group 4 in both 2010 and 2020, indicating ongoing challenges in this area. These countries are among the worst performers, indicating ongoing transport safety problems.

Over the past decade, there have been significant shifts in the transport safety groups in several countries. Brazil, for example, moved down from Group 2 to Group 4 in 2020, indicating a decline in safety performance. Chile also fell from Group 2 to Group 3 in 2020. However, Panama showed progress and moved up from Group 4 to Group 3 in 2020, indicating a shift in transport safety performance.

4.3 Robustness test

4.3.1 Comparison of the rankings

4.3.1.1 Initial sensitivity

The rankings of the countries for 2010 and 2020 using three different normalization methods (MinMax, Vector, and Z-Score) are listed in Table 3.

Table 3 Ranking comparisons using three different normalization methods.

Country	2010			2020		
	MinMax	Vector	Z-Score	MinMax	Vector	Z-Score
AG	10	22	20	14	30	30
AR	21	34	33	16	19	18
BB	4	13	12	9	26	22
BZ	31	35	35	23	25	24
BO	32	25	27	33	35	35
BR	11	6	8	2	5	5

CA	1	1	1	1	1	1
CL	17	12	14	6	6	7
CO	12	8	7	22	12	14
CR	7	3	5	19	14	11
CU	9	7	6	15	8	6
DM	14	20	22	30	29	29
DO	24	18	17	34	22	26
EC	33	26	26	17	16	16
SV	27	10	11	26	17	17
GD	13	23	21	24	28	28
GT	26	33	34	27	31	31
GY	35	21	19	20	11	12
HT	34	29	30	35	27	27
HN	22	17	18	18	23	25
JM	23	32	32	31	33	32
MX	28	30	28	12	15	15
NI	30	28	29	10	4	8
PA	5	2	3	21	18	19
PY	25	15	15	25	21	21
PE	20	19	24	28	32	33
LC	19	27	25	32	34	34
VC	2	4	4	4	13	13
KN	8	16	16	13	24	23
SR	6	9	9	11	9	9
BS	16	31	31	7	10	10
TT	15	14	13	8	7	4
US	3	5	2	3	2	2
UY	29	24	23	5	3	3
VE	18	11	10	29	20	20

Note: MM=MinMax; MA=Max; VE=Vector

As shown in Table 3, the rankings of different countries in 2010 and 2020 using three different normalization methods (MinMax, Vector, and Z-Score) show remarkable similarity, indicating the robustness of the proposed model in maintaining consistent relative rankings. For example, UY showed remarkable consistency, ranking 29th, 24th, and 23rd in 2010, and improved to fifth, third, and third in 2020 under the MinMax, Vector, and Z-Score methods, respectively. This stability of the rankings across different methods demonstrates the reliability of the ranking results based on the proposed model.

Line plots are presented (Fig. 2) to illustrate the trends and the influence of different normalization methods on country rankings, providing a clearer understanding of the sensitivity and robustness of the proposed model.

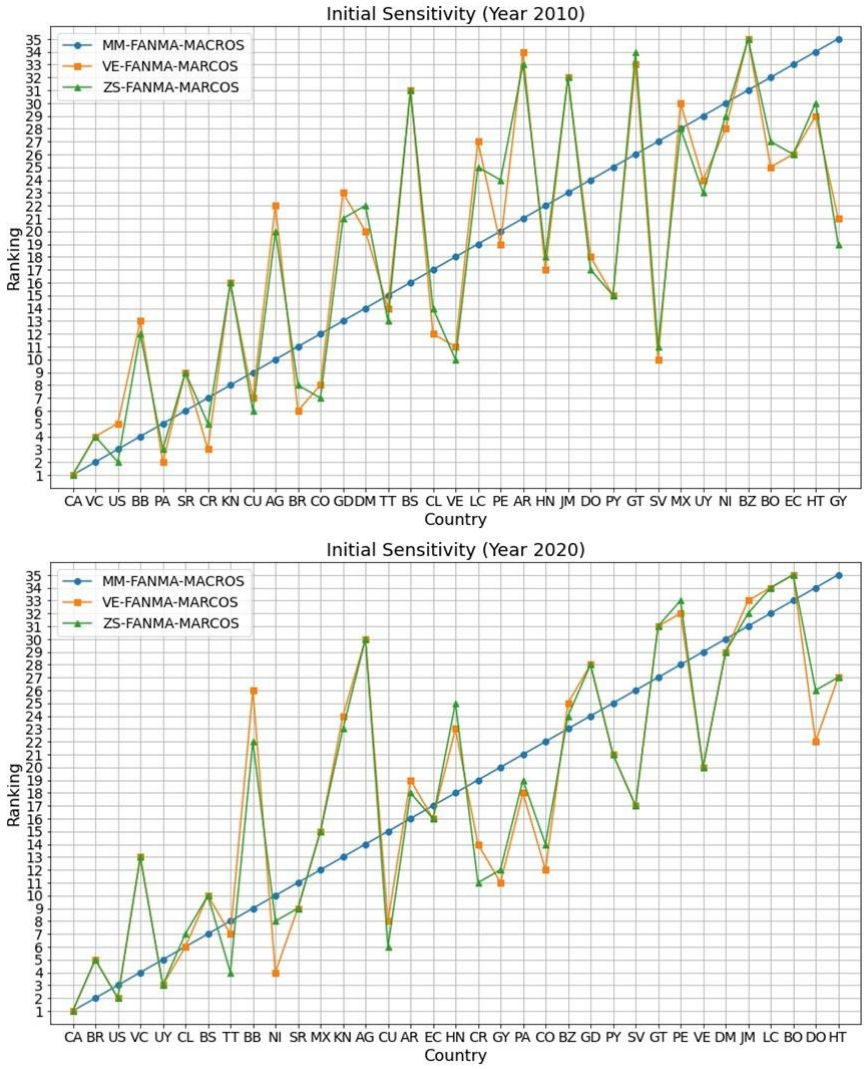


Fig. 2 Ranking comparisons between different normalization methods.

As shown in Fig. 2, the plots show a strong similarity in the rankings between the three techniques, indicating the robustness of the proposed model. Furthermore, the plots illustrate the sensitivity of country rankings to the choice of normalization method, highlighting the importance of selecting an appropriate MCDM method for accurate and reliable results.

To further quantify the similarity of results between the different normalization methods, the correlation matrix is presented in Table 4. Ranking similarity metrics are critical in decision-making, and conventional measures, such as Spearman's rank correlation, can be limiting. The weighted rank correlation coefficient improves this by assigning more importance to higher ranks, ensuring a more accurate similarity measure, especially when the top ranks are more significant [58]

Table 4 Weighted rank correlation coefficient of rankings across different normalization methods.

Methods	2010			2020		
	MinMax	Vector	Z-Score	MinMax	Vector	Z-Score
MinMax	1.00			1.00		
Vector	0.72	1.00		0.78	1.00	

21

Z-Score	0.74	0.99	1.00	0.81	0.99	1.00

As shown in Table 4, a moderate to high similarity between MinMax and Vector was observed in 2010, with a correlation of 0.72 and 0.74. In 2020, the correlation increased to 0.78 and 0.81, whereas the correlation between Vector and Z-Score remained high at 0.99. This indicates the robustness of the proposed model and thus increases the credibility of the analysis results.

4.3.1.2 Intermediate uncertainty

The ranks of the different countries for 2010 and 2020 using different weighting methods (FANMA, CRITIC, and Entropy) are listed in Table 5. These methods are applied to weight the criteria in the decision-making process, which affects the final rankings.

Table 5 Ranking comparisons under three different weighting methods.

Country	2010			2020		
	FANMA	CRITIC	Entropy	FANMA	CRITIC	Entropy
AG	10	10	10	14	11	13
AR	21	26	26	16	15	16
BB	4	4	4	9	3	5
BZ	31	34	33	23	16	18
BO	32	32	32	33	34	34
BR	11	11	12	2	4	3
CA	1	2	1	1	1	1
CL	17	18	18	6	7	6
CO	12	14	11	22	26	25
CR	7	6	6	19	21	22
CU	9	7	7	15	17	15
DM	14	17	17	30	27	31
DO	24	25	25	34	31	30
EC	33	30	31	17	12	11
SV	27	19	19	26	19	20
GD	13	12	13	24	25	26
GT	26	29	28	27	30	28
GY	35	35	35	20	23	21
HT	34	33	34	35	35	35
HN	22	22	22	18	20	19
JM	23	23	23	31	29	29
MX	28	31	30	12	18	17
NI	30	28	29	10	10	10
PA	5	5	5	21	24	23
PY	25	20	21	25	28	27
PE	20	24	24	28	33	33
LC	19	21	20	32	32	32
VC	2	1	2	4	6	7
KN	8	8	9	13	14	14
SR	6	9	8	11	13	12
BS	16	16	16	7	8	8
TT	15	15	15	8	9	9
US	3	3	3	3	2	2
UY	29	27	27	5	5	4
VE	18	13	14	29	22	24

Note: FM= FANMA; CT= CRITIC; EN= Entropy

22

As shown in Table 5, despite the different weighting methods, the rankings are very similar, indicating the robustness of the proposed model. For example, BR is close for all methods and has minimal effect on country positions under FANMA, CRITIC, and Entropy, underscoring the reliability of the outcomes.

Line plots are presented to further illustrate the influence of the weighting models on the country rankings. They show the changes between the different models and thus improve the understanding of the intermediate-phase uncertainty (Fig. 3).

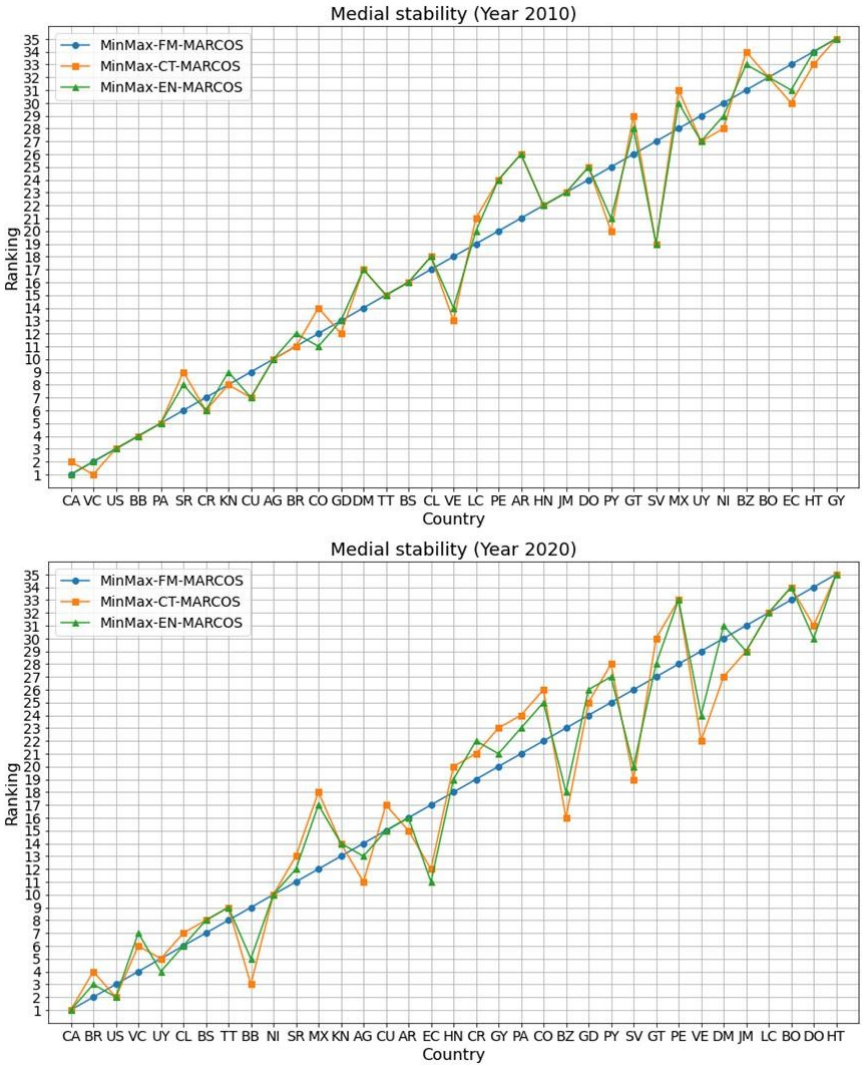


Fig. 3 Ranking comparisons between different weighting methods.

As shown in Fig. 3, the alignment of the lines for each method shows the consistency of the rankings across different weighting approaches, demonstrating the robustness of the proposed model. Despite the differences in the criteria weighting, overall ranking patterns remained stable, indicating confidence in the reliability of the results obtained with the proposed model.

The ranking similarity matrix was introduced to assess the similarity of the rankings produced using the different weighting methods (Table 6).

Table 6 Weighted rank correlation coefficient of rankings across different weighting methods.

Methods	2010			2020		
	FANMA	CRITIC	Entropy	FANMA	CRITIC	Entropy

23

FANMA	1.00			1.00		
CRITIC	0.97	1.00		0.95	1.00	
Entropy	0.97	1.00	1.00	0.96	0.99	1.00

As shown in Table 6, the correlation between FANMA and CRITIC and between FANMA and Entropy in 2010 was 0.97. The correlation between CRITIC and Entropy was 1.00, indicating almost identical results between these two models. In 2020, the correlation between FANMA and CRITIC was 0.95, and between FANMA and Entropy was 0.96. The correlation between CRITIC and Entropy remained high at 0.99. The high correlation values of the FANMA method confirmed its consistency and reliability compared with CRITIC and Entropy. The robustness of FANMA was improved across different models, making it the chosen weighting model for our primary analysis. This ensured stable and consistent rankings that effectively supported the decision-making processes.

4.3.1.3 Transverse reliability

The rankings of the different countries for 2010 and 2020 using different aggregation methods (MARCOS, TOPSIS [59], and VIKOR) are listed in Table 7. These methods were used to assess the transverse reliability of the rankings and to gain insight into the effect of the different decision-making models on the final rankings.

Table 7 Ranking comparisons under three different aggregating methods.

Country	2010			2020		
	MARCOS	TOPSIS	VIKOR	MARCOS	TOPSIS	VIKOR
AG	10	8	7	14	13	12
AR	21	21	19	16	15	15
BB	4	4	4	9	9	9
BZ	31	28	27	23	22	24
BO	32	32	32	33	31	29
BR	11	12	11	2	4	3
CA	1	2	1	1	1	1
CL	17	16	16	6	8	8
CO	12	13	14	22	21	19
CR	7	5	9	19	20	16
CU	9	10	15	15	18	22
DM	14	15	12	30	30	26
DO	24	19	21	34	34	33
EC	33	34	33	17	19	25
SV	27	31	30	26	27	32
GD	13	11	10	24	24	21
GT	26	24	25	27	26	27
GY	35	35	35	20	23	20
HT	34	33	34	35	35	35
HN	22	23	26	18	16	17
JM	23	18	20	31	29	31
MX	28	27	23	12	14	11
NI	30	30	31	10	12	14
PA	5	7	8	21	17	18
PY	25	26	28	25	25	23
PE	20	22	24	28	28	28

24

LC	19	20	17	32	33	30
VC	2	1	2	4	3	4
KN	8	9	6	13	11	13
SR	6	6	5	11	10	10
BS	16	17	13	7	5	5
TT	15	14	18	8	7	7
US	3	3	3	3	2	2
UY	29	29	22	5	6	6
VE	18	25	29	29	32	34

Note: MS= MARCOS; TP= TOPSIS; VK= VIKOR

As shown in Table 7, the results show a strong consistency between the different aggregation methods, indicating the robustness of the proposed model. Canada ranked first in all methods in both years, whereas the US remained stable within the top three places and ranked third in each method. This consistency emphasizes the reliability of the overall trends in the rankings.

Line plots are presented in Fig. 4 that visually depict the changes in country rankings across different aggregation methods to improve the understanding of transverse reliability.

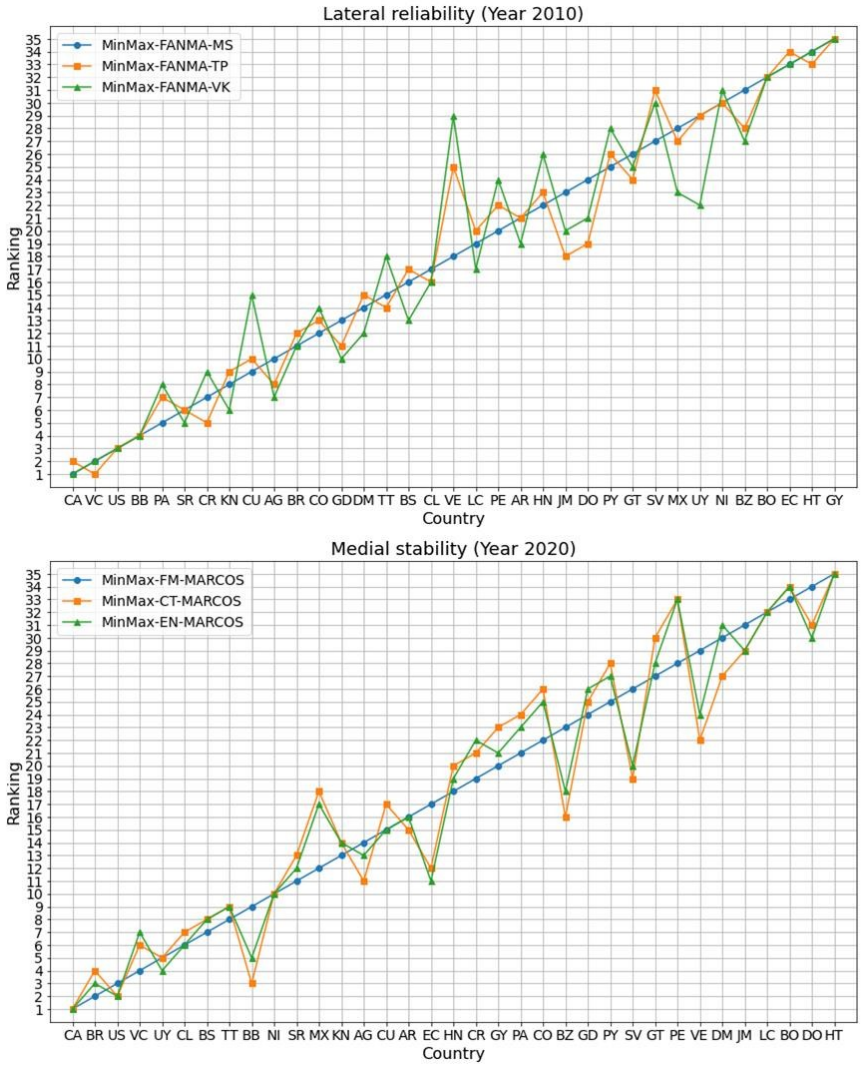


Fig. 4 Ranking comparisons across different aggregation methods.

25

As shown in Fig. 4, these line plots show that MARCOS, TOPSIS, and VIKOR have similar ranking patterns, indicating the reliability of the ranking results.

The ranking similarity matrix (Table 8) provides a detailed correlation analysis between the methods, which emphasizes the robustness of our chosen approach.

Table 8 Weighted rank correlation coefficient of rankings across different aggregating methods.

Methods	2010			2020		
	MARCOS	TOPSIS	VIKOR	MARCOS	TOPSIS	VIKOR
MARCOS	1.00			1.00		
TOPSIS	0.98	1.00		0.99	1.00	
VIKOR	0.95	0.97	1.00	0.96	0.97	1.00

As shown in Table 8, a strong correlation between MARCOS and TOPSIS was observed in 2010, with a high similarity of 0.98 and 0.95, respectively. The correlation between TOPSIS and VIKOR was also high at 0.97. The correlation values remain high in 2020, confirming the consistency and reliability of the ranking results.

MARCOS was used as the aggregation method for the main analysis, as it shows a strong correlation with other classical established methods. This ensures robust and reliable outcomes and makes the model well-suited for different decision-making scenarios. This choice increases the validity of the approach and supports the consistency of the results, ensuring the reliability of the model.

4.3.1.4 External comparability

The external comparability of the proposed model and the official HDI is presented in Table 9.

Table 9 Ranking comparisons under three different measurement methods.

Country	2010		2020	
	MinMax–FANMA–MARCOS	HDI	MinMax–FANMA–MARCOS	HDI
AG	10	6	14	11
AR	21	3	16	4
BB	4	8	9	11
BZ	31	26	23	30
BO	32	29	33	29
BR	11	22	2	17
CA	1	1	1	1
CL	17	4	6	3
CO	12	19	22	18
CR	7	13	19	7
CU	9	11	15	13
DM	14	24	30	25
DO	24	25	34	15
EC	33	17	17	22
SV	27	30	26	31
GD	13	10	24	10
GT	26	33	27	33
GY	35	31	20	26
HT	34	35	35	35
HN	22	34	18	34

26

JM	23	27	31	27
MX	28	16	12	18
NI	30	32	10	32
PA	5	12	21	9
PY	25	28	25	23
PE	20	21	28	16
LC	19	18	32	24
VC	2	19	4	18
KN	8	14	13	14
SR	6	22	11	21
BS	16	5	7	8
TT	15	6	8	6
US	3	1	3	2
UY	29	9	5	5
VE	18	15	29	28

Table 9 shows a strong correlation between the proposed model and the HDI, which underscores the validity of the model. For example, Canada consistently ranked first in both the proposed model and the HDI for both years, indicating a strong correspondence with official measures such as the HDI and thus reinforcing the credibility of the proposed model.

To better understand these comparisons, line plots were used to illustrate the changes and similarities in rankings between the proposed model and HDI (Fig. 5), providing a comprehensive overview of the evolution of the rankings based on the two approaches.

27

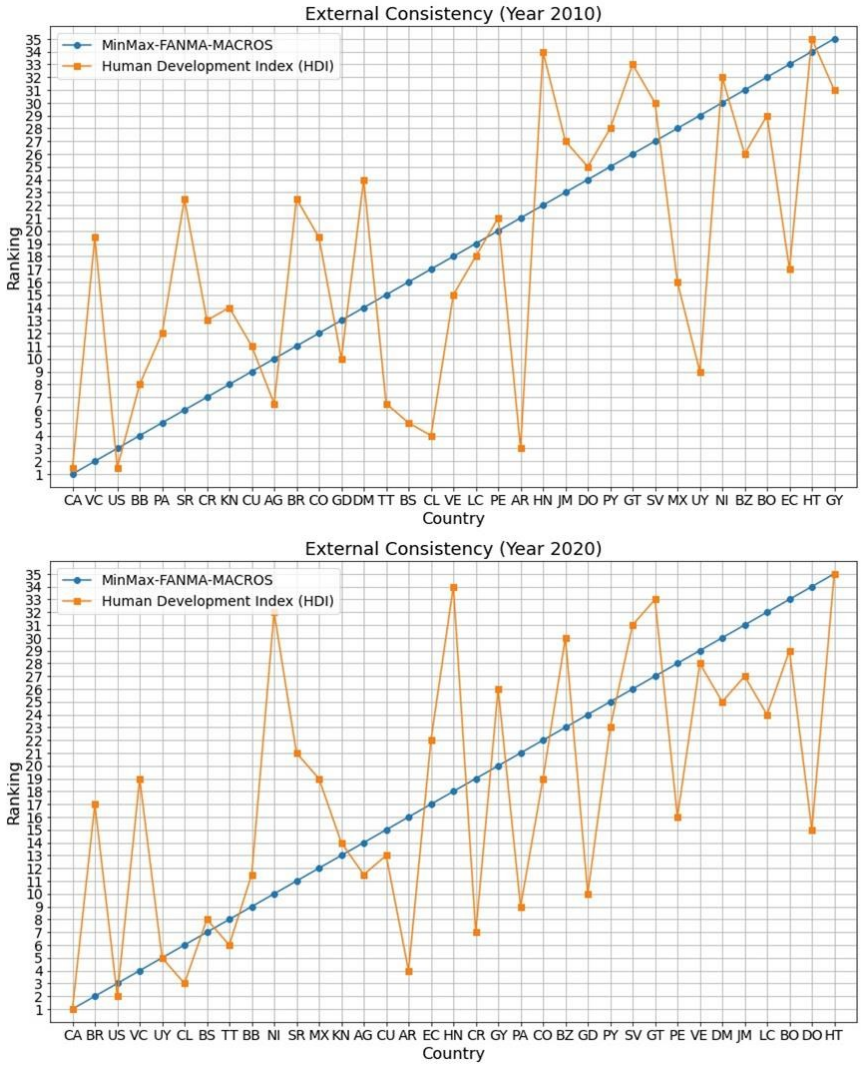


Fig. 5 Ranking comparisons between different measurement methods.

As shown in Fig. 5, the proposed model and HDI consistently produced comparable trends despite some differences. The rankings of the proposed model closely followed HDI rankings for many countries, particularly top-ranked countries, such as the United States and Canada. This correspondence with the HDI increased the external validity of the model.

To quantify the degree of correlation between the rankings of the proposed model and the HDI, similarity metrics were introduced (Table 10). These provide an accurate measure of their correlation and help to assess the robustness and validity of the proposed model compared to the established HDI, thereby presenting a clearer understanding of their agreement.

Table 10 Weighted rank correlation coefficient of rankings across different measurement methods.

Methods	2010		2020	
	MinMax–FANMA–MARCOS	HDI	MinMax–FANMA–MARCOS	HDI
MinMax–FANMA–MARCOS	1.00	0.61	1.00	0.61

28

HDI	0.61		1.00	0.61		1.00

As shown in Table 10, the proposed model, which aligned with the HDI, has a moderate positive correlation coefficient of 0.61 for both years, suggesting its robustness in providing a comparable yet distinct perspective on the development rankings, despite some differences in the rankings due to unique aspects of development.

4.3.2 Comparison of the grouping

4.3.2.2 Transverse reliability

The results of the different grouping methods—K-means with PSO, conventional K-means, and hierarchical cluster analysis (HCA)—for 2010 and 2020 are listed in Table 11. The cluster numbers indicate the performance levels, with lower numbers indicating better performance (1 being the best and 4 being the worst).

Table 11 Grouping comparisons under three different clustering methods.

Country	2010			2020		
	K-means with PSO	K-means	HCA	K-means with PSO	K-means	HCA
AG	2	2	3	3	3	3
AR	2	2	2	4	4	4
BB	2	2	3	3	3	3
BZ	4	4	4	4	4	4
BO	4	4	4	4	4	4
BR	2	2	2	4	4	4
CA	1	1	1	1	1	1
CL	2	2	3	3	3	3
CO	4	4	2	4	4	4
CR	4	4	2	3	3	3
CU	2	2	2	4	4	4
DM	4	4	2	4	4	4
DO	4	4	4	4	4	4
EC	4	4	4	4	4	4
SV	4	4	4	4	4	4
GD	4	4	2	4	4	4
GT	4	4	4	4	4	4
GY	4	4	4	4	4	4
HT	4	4	4	4	4	4
HN	4	4	4	4	4	4
JM	4	4	4	4	4	4
MX	2	2	2	4	4	4
NI	4	4	4	4	4	4
PA	4	4	2	3	3	3
PY	4	4	4	4	4	4
PE	4	4	4	4	4	4
LC	4	4	2	4	4	4
VC	4	4	2	4	4	4
KN	2	2	3	3	3	3
SR	4	4	2	4	4	4
BS	3	3	3	3	3	3
TT	2	2	3	3	3	3
US	1	1	1	2	2	2

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UY	2	2	3	3	3	3
VE	2	2	3	4	4	4

Note: HCA= Hierarchical Cluster Analysis

As shown in Table 11, countries such as Canada and the United States consistently showed high performance in Cluster 1 for all three approaches, indicating the robustness of the results, despite the differences in the clustering algorithms, which highlights the consistency between the different grouping techniques.

The PSO-enhanced K-means approach was evaluated using three key metrics: Silhouette Score, Davies-Bouldin Index, and Calinski-Harabasz Index, to assess the effectiveness and reliability of the models compared to the conventional methods and provide a comprehensive assessment of clustering quality (Table 12).

Table 12 Assessment of the clustering quality of the three clustering methods.

Method	2010			2020		
	SS	DB	CH	SS	DB	CH
K-means with PSO	0.597	0.319	212.755	0.648	0.280	213.406
K-means	0.597	0.319	212.755	0.648	0.280	213.406
HCA	0.563	0.430	193.648	0.648	0.280	213.406

Note: SS = Silhouette Score, DB = Davies-Bouldin Index, CH = Calinski-Harabasz Index

As shown in Table 12, **Silhouette Score** [60], a measure of the similarity of an object to its own cluster, ranges from -1 to 1, with higher values indicating better-defined clusters. In 2010, K-means with PSO and conventional K-means achieved a Silhouette Score of 0.597, whereas HCA achieved 0.563. In 2020, all methods achieved a score of 0.648, indicating improved clustering quality.

Davies-Bouldin Index (DB): Introduced by Davies and Bouldin [61], measures the average similarity ratio of each cluster to the most similar cluster, with lower values indicating better clustering quality. In 2010, K-means with PSO and conventional K-means had a DB index of 0.319, whereas HCA had a higher index of 0.430. By 2020, all methods achieved a lower DB index of 0.280, reflecting improved clustering performance.

Calinski–Harabasz Index (CH): Developed by Caliński and Harabasz [62], also known as the variance ratio criterion, measures the ratio of between-cluster dispersion to within-cluster dispersion. Higher values indicate better clustering. In 2010, K-means with PSO and conventional K-means achieved a CH index of 212.755, higher than that of HCA at 193.648. In 2020, both K-means achieved a CH index of 213.406, further confirming their superior clustering quality.

The PSO-enhanced K-means method outperformed conventional K-means and HCA, with consistently high Silhouette scores and Calinski-Harabasz indexes, and low Davies-Bouldin indexes, indicating well-defined and high-quality clusters.

4.3.2.3 External comparability

The performance levels of the K-means with PSO compared to the HDI external official results for 2010 and 2020 are presented in Table 13, with lower numbers indicating

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800 better performance. It also showcases the external comparability of the PSO-K-means
801 method with the established HDI.

802 Table 13 Grouping comparisons under different measurement indices.

Country	2010		2020	
	K-means with PSO	HDI	K-means with PSO	HDI
AG	2	2	3	2
AR	2	1	4	1
BB	2	2	3	2
BZ	4	2	4	3
BO	4	3	4	3
BR	2	2	4	2
CA	1	1	1	1
CL	2	1	3	1
CO	4	2	4	2
CR	4	2	3	1
CU	2	2	4	2
DM	4	2	4	2
DO	4	2	4	2
EC	4	2	4	2
SV	4	3	4	3
GD	4	2	4	2
GT	4	3	4	3
GY	4	3	4	2
HT	4	4	4	4
HN	4	3	4	3
JM	4	2	4	2
MX	2	2	4	2
NI	4	3	4	3
PA	4	2	3	1
PY	4	3	4	2
PE	4	2	4	2
LC	4	2	4	2
VC	4	2	4	2
KN	2	2	3	2
SR	4	2	4	2
BS	3	1	3	1
TT	2	2	3	1
US	1	1	2	1
UY	2	2	3	1
VE	2	2	4	3

803 *Note*: HDI=Human Development Index

804 As shown in Table 13, the PSO-enhanced K-means method is comparable to the HDI,
805 with countries, such as HT and BZ, consistently performing poorly. Similarly, NI
806 remained in Group 4 under both methods, reflecting its low HDI rankings. This external
807 comparability enhances the validity of the PSO-enhanced K-means method as it provides a
808 reliable and consistent approach to grouping countries based on their performance levels.
809 This comparison with established indices, such as the HDI, increased the credibility and
810 applicability of our approach in real-world scenarios.

811 To further evaluate the similarity of groups using the proposed model and
812 HDI, two key metrics were used: adjusted rand index (ARI) and normalized mutual

information (NMI). The ARI and NMI metrics assessed the agreement and mutual dependence between two clustering results, with values ranging from -1 to 1, and 0 to 1, respectively, indicating perfect agreement in both metrics. They provided a comprehensive assessment of the clustering similarity between different measures (Table 14).

Table 14 Assessment of the clustering similarity between the two measures.

Method	2010		2020	
	ARI	NMI	ARI	NMI
K-means with PSO	1.000	1.000	1.000	1.000
HDI	0.135	0.373	0.225	0.334

Note: ARI = Adjusted Rand Index, NMI= Normalized Mutual Information.

As shown in Table 14, the PSO-enhanced K-means method had an ARI and NMI of 1.000 for 2010 and was used as the baseline. The HDI method had an ARI of 0.135 and an NMI of 0.373, indicating moderate similarity with the PSO enhanced K-means method. For 2020, the PSO-enhanced K-means method again had perfect ARI and NMI values. The HDI showed better agreement with the PSO-enhanced K-means method (ARI = 0.225 and NMI = 0.334). The moderate agreement with the HDI underscored the reliability and effectiveness of the K-means with PSO model in producing consistent and comparable clustering results with the well-known HDI.

5 Practical guidance

5.1 Dynamic changes in indicators

The dynamic changes in the performance of different countries on several criteria from 2010 to 2020 are shown in Fig. 6 using spider plots. Each axis of the spider plot represents a different country, with the distance from the center indicating the performance score for a specific criterion. These visualizations provide insights into the evolution of each country's performance over the decade, highlighting trends, such as improvement, deterioration, stability, or irregular changes. By examining these charts, it was possible to identify countries that have made significant progress, those that have remained stable, and those that have experienced fluctuating levels of performance.

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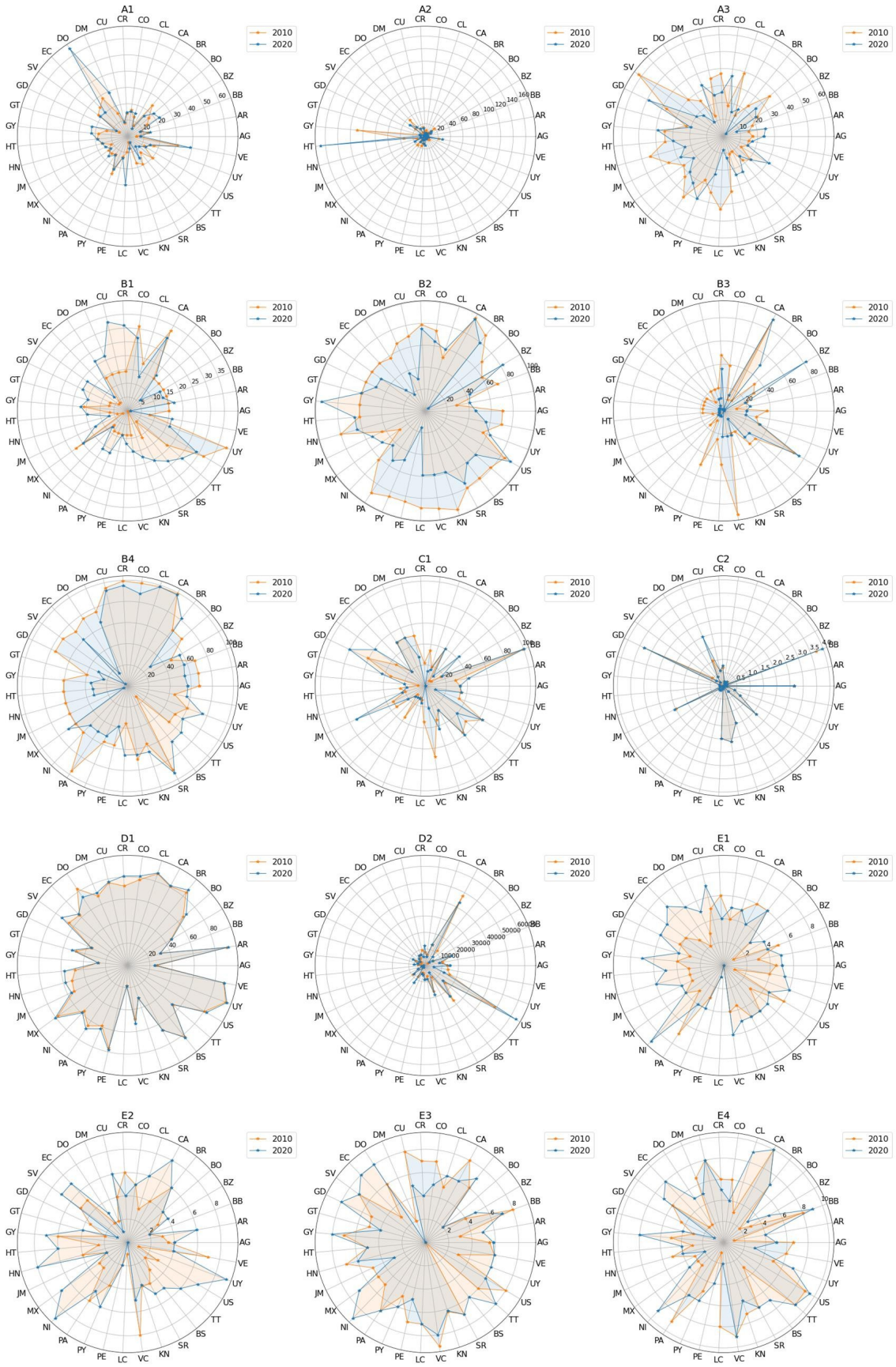


Fig. 6 Change trend of the SPIs.

(1) Criterion A1

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For criterion A1 (Road fatalities per 100,000 inhabitants), the spider plot indicates a general improvement in performance for most countries from 2010 to 2020, with Brazil and Colombia showing significant improvements, whereas Venezuela and Haiti show relatively stable performance with minor changes over the last decade.

(2) Criterion A2

Criterion A2 (Road fatalities per 10,000 registered vehicles) showed a mixed pattern. Barbados and Argentina improved significantly, whereas Ecuador and Guatemala have remained unchanged. Belize and Bolivia have made remarkable progress in this criterion.

(3) Criterion A3

For criterion A3 (Percentage of pedestrian fatalities out of the total fatalities (%)), most countries have improved their performance from 2010 to 2020. Brazil and Colombia have made significant progress. However, some countries, such as Haiti and Guyana, show fluctuations, indicating an irregular performance pattern.

(4) Criterion B1

The plot for criterion B1 (Road traffic deaths involving alcohol (%)) showed a general trend of improvement in many countries. Brazil and Argentina show notable progress, whereas countries, such as Venezuela and Nicaragua show minor changes, indicating relative stability in their performance.

(5) Criterion B2

Criterion B2 (Seatbelt wearing rates in front seats (%)) showed significant improvements in most countries. Canada and the US have made slight progress from 2010 to 2020, while Bolivia and Haiti have made substantial improvements and are moving well ahead.

(6) Criterion B3

For criterion B3 (Seatbelt wearing rates in rear seats (%)), most countries showed improved performance in 2020 compared to 2010. Brazil and Barbados demonstrated significant improvements, whereas Guyana and Haiti had mixed results, reflecting both improvements and declines over this period.

(7) Criterion B4

For criterion B4 (Helmet-wearing rates (%)), most countries improved from 2010 to 2020. Brazil and Argentina have improved significantly, whereas Haiti and Guyana remained relatively stable with little change.

(8) Criterion C1

There were mixed results for criterion C1 (Paved roads (% of total)). Barbados [24] and Belize showed notable improvements, whereas countries Guyana and Haiti showed

34

irregular patterns, with varying performance. Some countries, such as Canada and the United States, remained relatively stable.

(9) Criterion C2

For criterion C2 (Road network density (km/km²)), most countries improved significantly over the decade, particularly Barbados and Argentina. Conversely, Haiti and Guyana showed minimal change, indicating stable but lower performance levels.

(10) Criterion D1

Criterion D1 (Urban population (%)) showed a general trend of improvement in many countries. Brazil and Argentina made significant progress, whereas Venezuela and Nicaragua showed minor changes, indicating relative stability in their performance.

(11) Criterion D2

Criterion D2 (GDP per capita (current US$)) showed significant improvements in most countries, with Barbados and Brazil leading the way. However, Venezuela and Guyana have improved even more significantly and have moved well up the axis.

(12) Criterion E1

For criterion E1 (Enforcement score on speed limit law), most countries improved performance in 2020 compared to 2010. Brazil and have improved significantly, whereas Guyana and Haiti have seen both improvements and declines during this period.

(13) Criterion E2

For criterion E2 (Enforcement score on drink-driving law), the spider plot shows an overall improvement in most countries from 2010 to 2020, with Brazil and Barbados showing significant progress. However, Haiti and Guyana showed relatively stable performance with minor changes over the last decade.

(14) Criterion E3

For criterion E3 (Enforcement score on seatbelt law), the plot shows significant improvements in many countries, with Brazil and Argentina standing out with substantial improvements. Conversely, Haiti and Guyana showed irregular patterns, indicating fluctuating levels of performance between 2010 and 2020.

(15) Criterion E4

Criterion E4 (Enforcement score on helmet use law) shows a general trend toward improvement in numerous countries. Canada and the United States maintained high performance levels, with slight progress between 2010 and 2020. Brazil and Barbados improved significantly, whereas Venezuela showed both improvements and declines.

5.2 Deconstructing the composite index

The performance scores of the different countries are shown in Fig. 7, with each bar representing a composite score for each country. The colors within each bar indicate the contribution of each criterion to the overall score. The visualization deconstructs the composite index by identifying the most important criteria for each country's performance and provides insights into their strengths and focus areas.

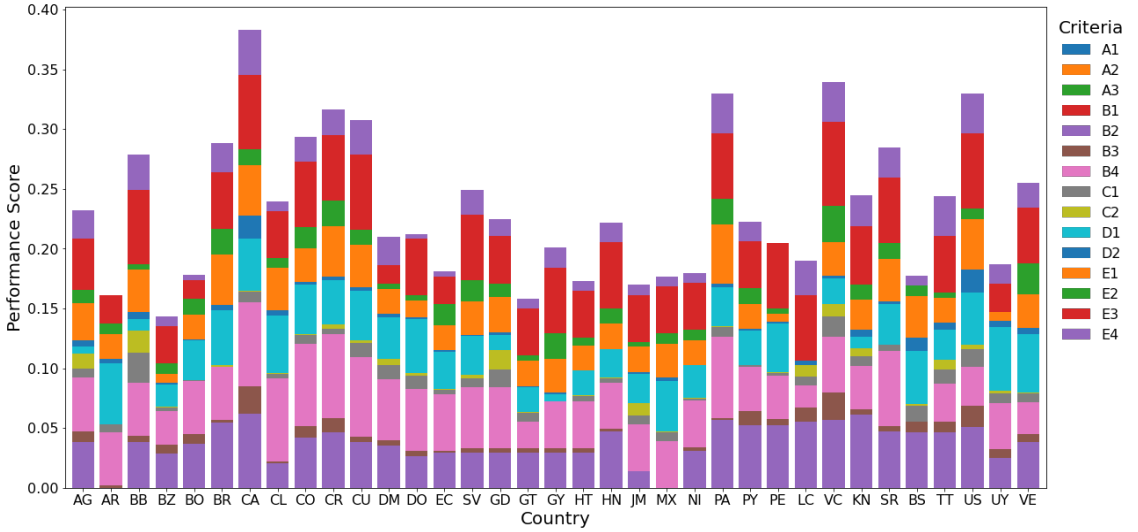


Fig. 7 Deconstruction of a composite score for each country.

(1) Canada

Canada scored the highest, with significant contributions from criteria B2 (seatbelt wearing rates in front seats), B3 (seatbelt wearing rates in rear seats), and E1 (enforcement score on speed limit laws). Th high score for B2 and B3 underscores Canada's strong commitment to transport safety measures, whereas Criterion E1 highlights the effective enforcement of traffic laws in Canada.

(2) Brazil

Brazil showed a strong performance, particularly in Criteria B1 (road traffic deaths involving alcohol), B3 (seatbelt wearing rates in rear seats), and E3 (enforcement score on seat belt law). Criteria B1 and B3 show Brazil's progress in improving transport safety and public health. Criterion E3 also contributes significantly to Brazil's overall score and reflects progress in enforcing safety regulations.

(3) Barbados

Barbados showed a balanced performance with notable contributions from criteria A3 (percentage of pedestrian fatalities out of the total fatalities), B2 (seatbelt wearing rates in front seats), and E1 (enforcement score on speed limit laws). The high score in A3 and B2 underscores Barbados' effective governance and commitment to transport safety. Furthermore, Criterion E1 stood out, showcasing Barbados' focus on maintaining high enforcement standards.

36

(4) Argentina

Argentina's score was driven by strong performances in criteria A1 (road fatalities per 100,000 inhabitants), B1 (road traffic deaths involving alcohol), and E2 (enforcement score on drink-driving laws). Criteria A1 and B1 showcased Argentina's commitment to reducing road fatalities and improving public health. Criterion E2 further highlighted the country's strict enforcement of drink-driving laws.

(5) United States

The United States scored highly on several criteria, with criteria B2 (seatbelt-wearing rates in front seats), B3 (seatbelt-wearing rates in rear seats), and D1 (urban population) contributing significantly. Criteria B2 and B3 contributed significantly, indicating the country's robust transport safety measures. Criterion D1 further underscored the high rate of urbanization in the United States, which correlates with improved access to safety resources and infrastructure.

(6) Mexico

Mexico stood out in Criteria B2 (seatbelt-wearing rates in front seats), B4 (helmet-wearing rates), and D2 (GDP per capita). Mexico's commitment to improving transport safety and reducing the number of road fatalities is reflected in its strong performance in B2 and B4. Criterion D2 also contributed significantly, reflecting Mexico's economic growth and improved public health measures.

(7) Venezuela

Venezuela's performance shows significant contributions from Criteria A2 (Road fatalities per 10,000 registered vehicles), B1 (Road traffic deaths involving alcohol), and E4 (Enforcement score on helmet-use law). Criteria A2 and B1 highlight areas where Venezuela has made significant progress in transport safety despite broader challenges. In addition, criterion E4 played a crucial role and underscores the importance of the enforcement of the helmet law for Venezuela's overall safety performance.

(8) Guyana

Guyana's score was influenced by Criteria B2 (seatbelt-wearing rates in front seats), B3 (seatbelt-wearing rates in rear seats), and C1 (paved roads). Criteria B2 and B3 highlight Guyana's commitment to improving transport safety and health outcomes, while criterion C1 indicates progress in improving road infrastructure.

5.3 Decomposition of score changes

The changes in performance scores for countries from 2010 to 2020 are shown in Fig. 8, where each bar represents the overall change in score for a country, and different colors indicate the contribution of each criterion to the change in score. Positive and negative contributions are shown above and below the horizontal axis, respectively. This

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representation helps to break down the changes in composite scores by highlighting the specific criteria that led to the improvements or declines in each country's performance.

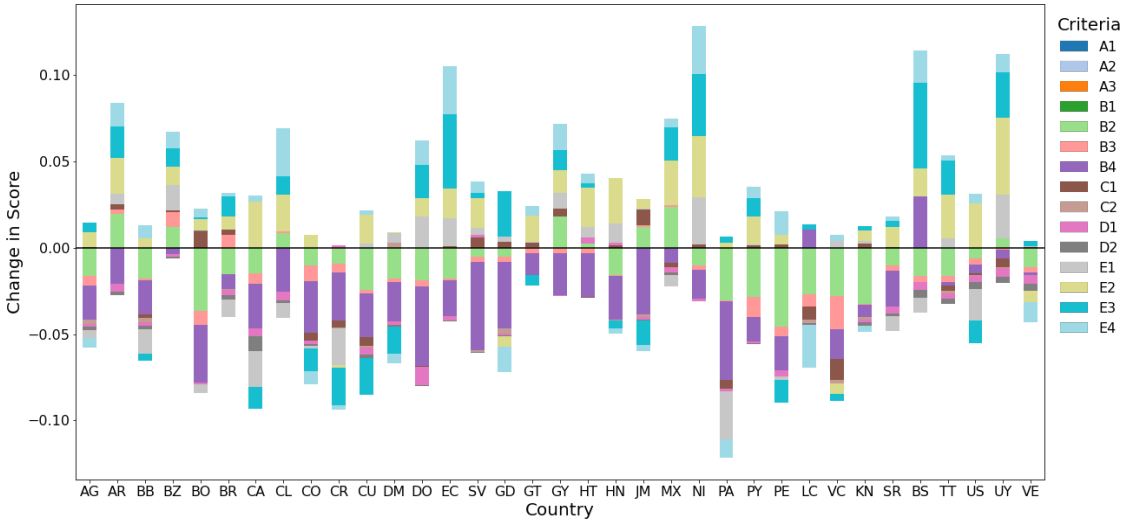


Fig. 8 Decomposition of changes in scores per country.

(1) Canada

The performance of Cananda showed positive changes, primarily driven by criteria B2 (seatbelt-wearing rates in front seats) and B3 (seatbelt-wearing rates in rear seats). These criteria contributed significantly to Canada's overall improvement and reflect progress in transport safety measures and enforcement.

(2) Brazil

Brazil showed positive changes, criteria B1 (road traffic deaths involving alcohol) and E3 (enforcement score on seatbelt law) contributing significantly. These improvements reflect Brazil's commitment to improving transport safety measures and reducing alcohol-related traffic deaths.

(3) Barbados

Barbados also achieved notable improvements, mainly resulting from Criteria A3 (percentage of pedestrian fatalities out of the total fatalities) and B2 (seatbelt-wearing rates in front seats). These criteria contributed significantly to the positive changes in the country's score and underscored the effective management of transport safety and economic stability.

(4) Argentina

Argentina's score changes were driven by positive contributions from Criteria A1 (road fatalities per 100,000 inhabitants) and E2 (enforcement score on drink-driving laws). These criteria underscored Argentina's commitment to maintaining transport safety and promoting drunk driving enforcement.

(5) United States

38

The United States showed mixed changes in transport safety performance, with both positive and negative contributions. Criteria B2 (seatbelt-wearing rates in front seats) and D1 (urban population) had a positive influence on the overall score, whereas declines in other areas showed a mixed trend in performance. This highlights the progress made in urban safety measures alongside challenges in other sectors.

(6) Mexico

Mexico's positive score changes were primarily driven by Criteria B2 (seatbelt-wearing rates in front seats) and D2 (GDP per capita). These criteria reflect Mexico's commitments to stabilize its economy and improve public safety, which contributed to an improvement in overall performance.

(7) Venezuela

Venezuela experienced a decline in performance with negative contributions from Criteria A2 (road fatalities per 10,000 registered vehicles) and B1 (road traffic deaths involving alcohol). These negative contributions underscored the Venezuela faces in certain areas, which ultimately affected its overall score. This highlights the urgent need for targeted action in the area of transport safety and public health in this country.

(8) Guyana

Guyana showed a mixed performance with both improvements and declines. The positive contributions from criterion B2 (seatbelt-wearing rates in front seats) and C1 (paved roads) indicate progress in economic and safety conditions, whereas other criteria had a negative effect on the overall score, showcasing both progress in infrastructure and continued challenges in other areas.

5.4 Benchmarking within each group

The grouping of the OAS countries into four distinct groups is shown in Fig. 9, with each group represented by a different color to illustrate the regional distribution of countries within each group. This map visually represents the grouping of countries based on their overall performance on various criteria.

Following the map, the spider plots in Fig. 10 show the performance of the different groups in relation to different criteria. Each axis of the spider plot represents a different criterion, with the distance from the center indicating the performance score for that criterion. These plots provide a visual representation of the strengths and weaknesses of each group in relation to the criterion and provide valuable insights into the countries that perform particularly well in certain areas and areas requiring improvement.

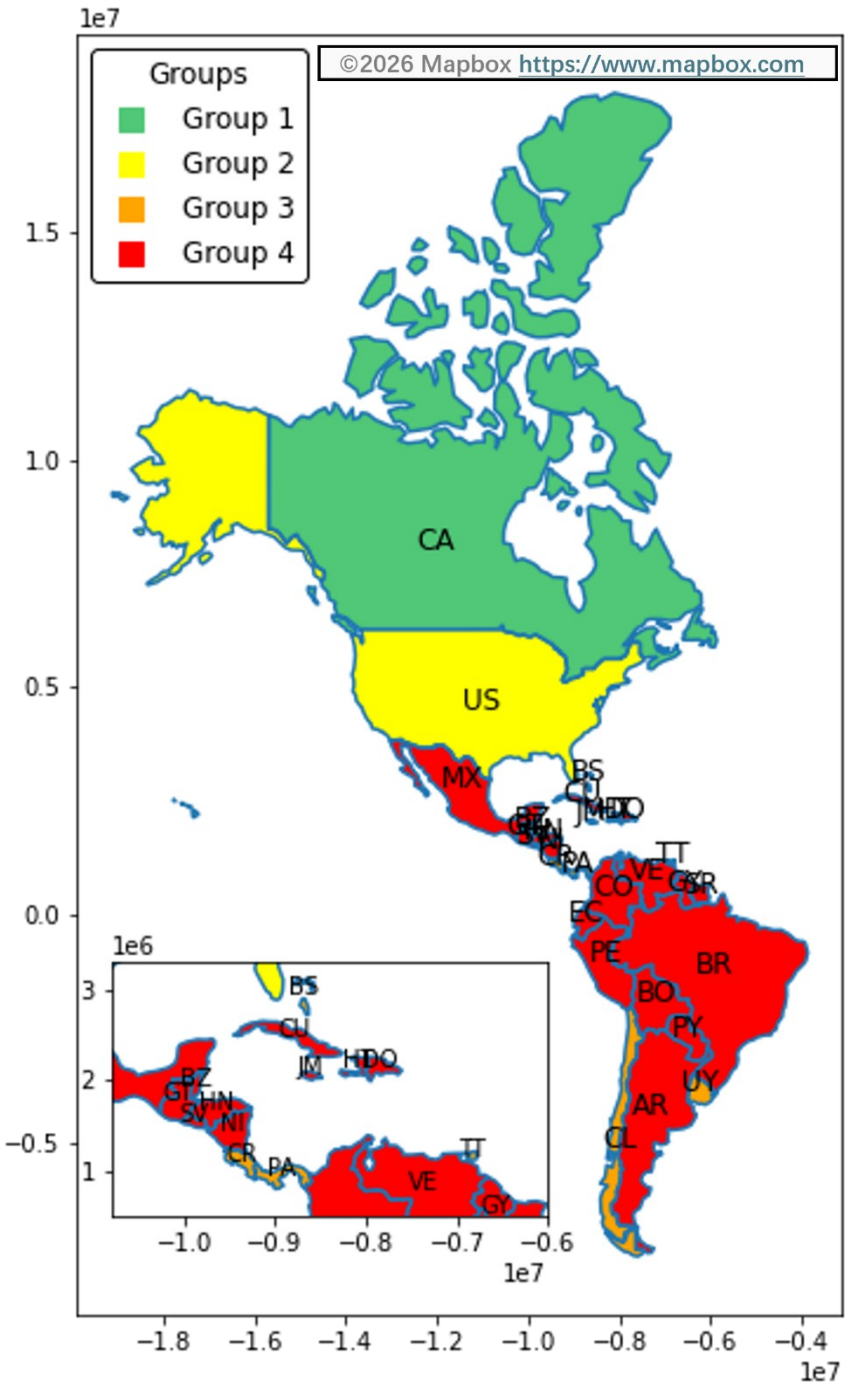


Fig. 9 Geographical distribution of OAS countries in terms of transport safety performance.

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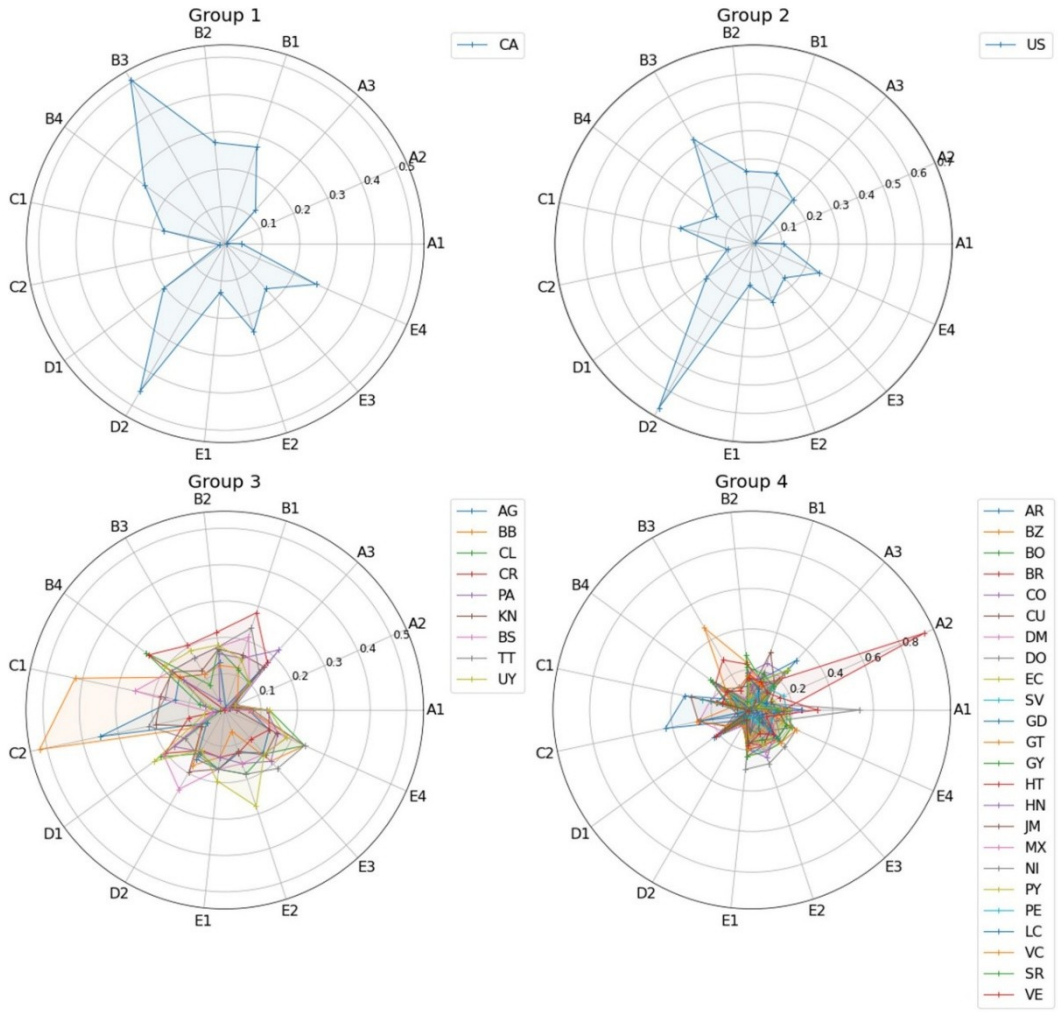


Fig. 10 Benchmarking within each group.

(1) **Group 1** consists exclusively of Canada:

- Canada exhibits exceptional performance in Criteria B2 (seatbelt-wearing rates in front seats), B3 (seatbelt-wearing rates in rear seats), and E1 (enforcement score on speed-limit law). These scores underscore Canada's comprehensive transport safety measures and strict enforcement of traffic laws.
- Canada stood out as a model within Group 1, particularly in terms of transport safety enforcement and administration.

(2) **Group 2** consists exclusively of the United States:

- The United States excelled in Criteria B2 (seatbelt-wearing rates in front seats), B3 (seatbelt-wearing rates in rear seats), and D1 (urban population), reflecting robust transport safety measures, high seatbelt compliance, and extensive urbanization.
- The United States served as a benchmark within Group 2, which can be attributed to its comprehensive approach to transport safety and urban development.

(3) **Group 3** included Argentina, Brazil, and Chile.

41

- Argentina exceeds in criteria A1 (road fatalities per 100,000 inhabitants) and A2 (road fatalities per 10,000 registered vehicles), indicating effective measures to reduce the number of road fatalities.
- Brazil excelled in criteria B1 (road traffic deaths involving alcohol) and E3 (enforcement score on seatbelt law), indicating significant progress in public health and safety enforcement.
- Chile outperforms in criteria C1 (paved roads) and D2 (GDP per capita), showcasing its robust infrastructure and economic development.
- These Group 3 countries can serve as a benchmark for others, offering best practices in transport safety, public health, and infrastructure development.

(4) **Group 4** comprises many countries, including Argentina, Brazil, and Venezuela.

- Both Argentina and Brazil showed consistent performance on several criteria. Argentina performed best on criteria A1 (road fatalities per 100,000 inhabitants) and E2 (enforcement score on drink-driving laws), whereas Brazil excelled in Criteria B1 (road traffic deaths involving alcohol) and E3 (enforcement score on seatbelt law).
- Despite facing significant challenges, Venezuela performed relatively better on criterion E4 (enforcement score on helmet use law).
- Argentina and Brazil serve as exemplary models for transport safety and law enforcement and provide valuable lessons for other countries in this group to emulate.

These groupings and benchmarks serve to highlight the best practices within each group and provide a framework for countries to learn from their peers and implement effective strategies to improve their overall performance. By examining the leading practices of the top-performing countries, other nations can adopt and adapt these strategies to address their own challenges and opportunities.

6 Conclusion and further studies

6.1 Concluding remarks

In this study, an innovative hybrid MCDM model, FANMA–MARCOS–K-means with PSO, was developed to support transport safety decision-making and policy formulation with a high degree of reliability and stability. This methodology will provide policymakers, decision-makers, and practitioners with a comprehensive understanding of the strengths and weaknesses in transport safety development and enable them in identifying benchmark practices and areas for improvement within their respective countries. It will serve as a valuable tool for strategic planning and policy making. Intensive comparisons of the empirical results using a real case study of the OAS countries showcased the robustness of the proposed model and highlighted its efficiency and applicability in real-world scenarios.

This study makes significant contributions to the literature and real-world scenarios.

42

(1) A comprehensive SPIs system for measuring and benchmarking transport safety development at the regional level was introduced, providing government officials, managers, and practitioners with a basic framework for making informed decisions and formulating targeted strategies.

(2) Our improvements to the K-means algorithm through the integration of PSO have addressed several key shortcomings. Conventional K-means algorithms often struggle to determine the optimal number of clusters and can be sensitive to the initial centroid selection. By incorporating PSO, the algorithm's ability to determine the optimal clustering solution was improved, its sensitivity to initial conditions reduced, and overall stability was increased. This improvement is particularly important when dealing with large datasets, making the method well-suited for real-world scenarios. The enhanced K-means algorithm ensured that the formation of clusters is based on actual data patterns rather than arbitrary initial decisions, leading to more meaningful and reliable outcomes.

(3) The proposed methodology provided countries in the OAS region a comprehensive evaluation tool to improve the planning, management, and enforcement of transport safety commitments. It enabled policymakers to assess key aspects, such as infrastructure, vehicle safety, and operational efficiency, identifying benchmark practices and areas for improvement. By using this tool, countries in the OAS region can work toward improving their coordination, leadership, and strategic planning, ultimately leading to significant progress in overall transportation safety across regions.

6.2 Limitations and future prospects

Despite the significant contributions of this study, several limitations must be acknowledged. Firstly, there were issues with data availability, which affected the comprehensiveness of the analysis. The absence of data from certain sources limited our ability to fully examine all criteria and countries. Secondly, the use of specific data normalization methods in weighting, as specified by MARCOS, limited the flexibility of our approach. This constraint prevented the selection of alternative data normalization methods that would have been more appropriate for specific contexts or improved the robustness of the weighting method.

To address these limitations and further advance the research, several future research directions are proposed. First, the current SPIs system should be updated to reflect recent developments in transportation scenarios, including cooperative vehicle infrastructure and autonomous driving. The inclusion of more relevant and updated indices will ensure that the evaluation framework remains relevant and comprehensive. Secondly, the inclusion of advanced algorithms can significantly improve the efficiency, stability, and running speed of the model, given the constant advances in the field of artificial intelligence. Specific methods, such as deep reinforcement learning and genetic algorithms, can be integrated to optimize the decision-making process. In addition, the use of supervised learning techniques, such as support vector machines or random

43

forests, can improve the predictive accuracy and adaptability of the model. Third, to increase the generalizability and practicality of the model, it is essential to extend its application beyond the OAS region to a global scale. This expansion would validate the applicability of the model in different contexts and improve its operational capabilities.

Author contributions

Shijie Sun: Data curation, Formal Analysis, Methodology, Writing – original draft. **Xinyue Zhang, Jialin Li, Kejun Lu:** Methodology, Software, Visualization, Writing – original draft. **Anna Qian, Zhanheng Sun, Mingren Zheng:** Investigation, Validation, Writing – original draft. **Faan Chen:** Conceptualization, Funding acquisition, Project administration, Supervision, Writing – review & editing.

Data availability

Data will be made available upon request.

Declaration of Competing Interest

The authors declare no competing interests.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

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45

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48